

EUV Lithography: State-of-the-Art Review

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Abstract: Although several years delayed than its initial plan, extreme UV lithography (EUVL) with 13.5nm wavelength has been finally implemented into high volume manufacture (HVM) of mainstream semiconductor industry since 2018. With the delivery and installation of ASML EUV scanners in those giant Fab players like Samsung, TSMC and Intel, EUV lithography is becoming a sort of industry standard exposure metrology for those critical layers of advanced technology nodes beyond 7nm. Although ASML NXE EUVL scanner is the only commercialized EUV exposure system available on the market, its development is the concentration of all essence from worldwide industrial and academic collaboration. It is becoming more and more important not only for fab runners but also for main stream fabless design houses to understand and participate the progress of EUVL. In this review, working principles, module structures and technical challenges have been briefly discussed regarding each EUV subsystem, including light source, reflection mirrors and system, reticle module as well as photoresist development. EUV specific issues of light intensity, defectivity within reflection system, line edge roughness (LER) and mask 3D effects have been focused respectively and promising solutions have been summarized as well.

Keywords: EUV lithography, EUV review, mask 3D, line edge roughness, EUV light source.

1. Introduction

The development and growth of semiconductor industry has been following Moore's law in recent 50 years, and it is fair to say that the continuous shrinking of transistors in microchips has been mainly driven by the development of photolithography technology^[1-3]. Since the appearance of the first G-line (436nm wavelength) step-and-repeat system, namely wafer stepper, in late 1970s, this exposure system quickly has become mainstream and dramatically promoted the capability and efficiency of photolithography^[4,5]. Due to the Rayleigh diffraction effect, smaller wavelength of light has to be adopted in order to get a reduced feature size with high resolution through a lens system, which makes a semiconductor device smaller. After jumping onto the fast lane of step-and-repeat system, the light source adopted in photolithography has been continuing switching from excimer laser for i-line (365nm)^[6], to krypton fluoride (KrF) excimer laser for 248nm deep ultraviolet^[7] and to argon fluoride (ArF) for 193nm wavelength in order to get even smaller feature size^[8,9]. On this track, 157nm wavelength of F₂ excimer laser was supposed to be produced to reach smaller half pitch than 65nm^[10,11]. However, wavelength is not the only factor to improve the resolution of smaller pitches. Numerical aperture, NA as the Acronym, plays another role by

the Rayleigh criterion:

$$R (\text{smallest feature size}) = k_1 * \lambda / NA$$

where λ is the wavelength of the light used, NA is the numerical aperture of the system's lens and k_1 is known as the resolution factor and accounts for all other process variables. Furthermore, since $NA = n * \sin \alpha$, where α is the light incident angle and n is the refractive index of the medium surrounding the lens, and liquid has a larger n than air, an immersion ArF scanner was developed to reach a higher resolution for smaller features, in which immersion fluid (mainly water) is well controlled to stay between last lens and wafer surface in order to get higher NA^[12] (Figure 1). The first commercial tool with this innovative technology was built and shipped by ASML in 2006 which started to dominate the market since then. The cost-effective implementation of immersion ArF scanner in the industry killed further development of tools with 157nm wavelength.

The implementation of immersion ArF scanner, enhanced by RET (Resolution Enhancement Technology) and multi-patterning technologies enabled the industry to reach 10nm/7nm FinFET tech node with acceptable cost, efficiency and die yield. But what is next? To resolve lines (half-pitch) smaller than 7nm, quadruple patterning or even more masks need to be used only for one functional layer, which will dramatically increase the cost and process

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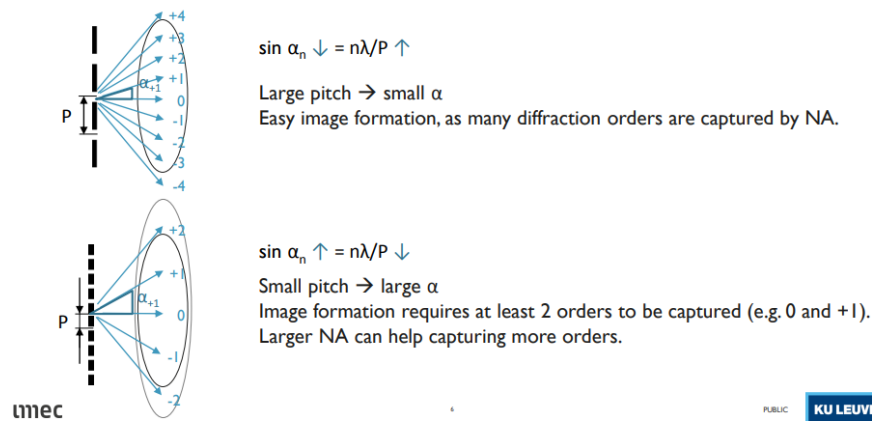


Figure 1. Illustration of Rayleigh criterion. Larger NA benefits smaller pitches. Courtesy of Vu Luong, IMEC^[14].

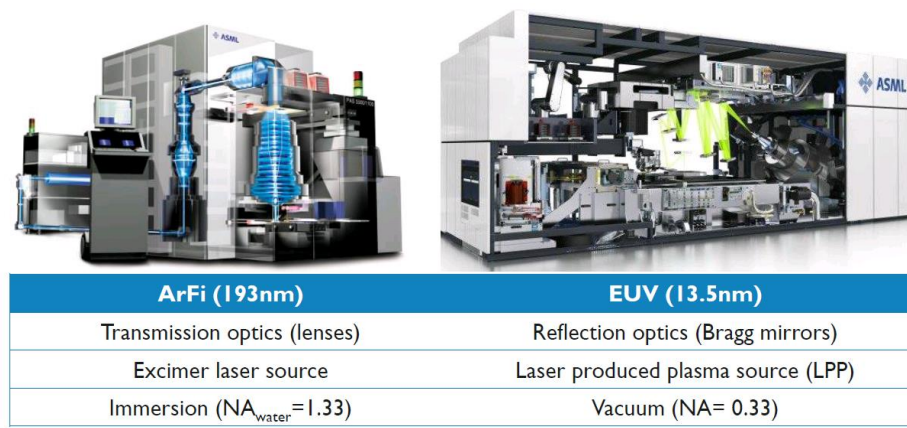


Figure 2. Comparison of an ASML ArF immersion scanner and an ASML EUV scanner. Courtesy of Vu Luong, IMEC^[14].

variation. The industry has to follow Rayleigh criterion again to push the exposure wavelength to the limit.

The industrialization of EUV lithography (EUVL) has been turned on since the turn of the 21st century and even earlier. Almost all key players from the industry and academic institutes have been involved in EUV light source development, Bragg mirrors with multi-coated layer, reflection system in vacuum, reflection reticle design and material development, EUV photoresist synthesis and optimization, contamination and defects reduction, DTCO (Design Technology Co-Optimization) and EDA tool development, as well as many other related engineering fields^[13]. Most of the accomplishments of research and development have been concentrated to the ASML NXE EUVL module, which began its journey of High Volume Manufacture (HVM)^[1,14] since 2017. Modern EUV scanner is equipped with a high power EUV light source which can generate 13.5nm wavelength radiation at a high power up to 250W (above 400W light source is also under development), a full reflection mirror system in a

vacuum chamber which prevent high adsorption on EUV optical path and maximize reflectivity, as well as redesigned reflection mirror and reticle system with 0.33 NA from Zeiss SMT to enable up to 140 wph (Wafer Per Hour) throughput. Figure 2 shows visual structures of a deep UV immersion module TWINSCAN NXT:1980i and a latest TWINSCAN NXE-3400B EUV module which is clearly distinguishable from the deep UV module due to the light reflection system. With the help of EUV lithography system, together with economic affordable multi-patterning technologies semiconductor technology node can be pushed further to beyond 3nm, and extend Moore's Law to next decades (Figure 3). In the following chapter, main sub-modules of a EUV scanner will be briefly reviewed.

2. Light Source

Unlike the direct deep UV generation from excimer lasers, extreme UV light with 13.5nm wavelength can only be produced by excitation and

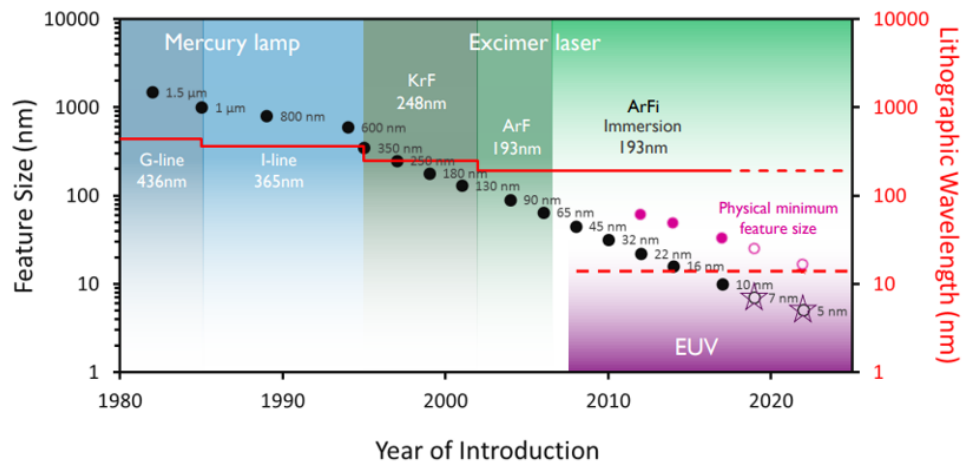


Figure 3. Evolution of patterning resolution and lithographic wavelength. Courtesy of Vu Luong, IMEC^[14].

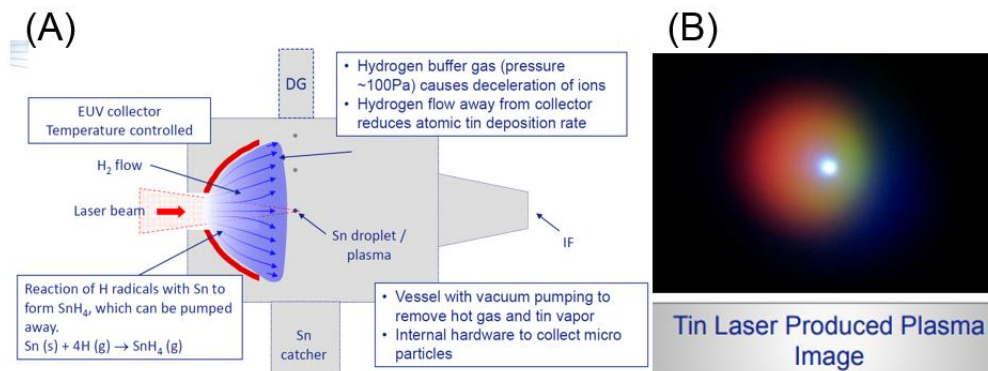


Figure 4. (A) Illustration of EUV light source vessel with hydrogen gas as buffer. (B) An image of laser produced plasma around a tin droplet^[21]. Courtesy of Igor Fomenkov, ASML.

relaxation of relatively inner electrons of atoms (e.g. Sn^{20+}), which could only be realized by atom ionization to form hot dense plasma^[15-17]. In a commercial ASML EUV system, high power CO_2 laser with 40~60kW has been applied to bombard liquid tin (Sn) droplet to generate dense plasma at high temperature around the droplet, where EUV radiation takes place^[18,19]. EUV photons are collected by a multilayer coated collector in an ellipsoidal shape (Figure 4A) and conducted to the reflection optics in a scanner. As a standalone component of ASML EUV scanner, the LPP (laser produced plasma) EUV light source is composed of high power laser generator, laser beam delivery system and EUV source vessel, which is one of the most distinguish parts of a EUV system compared with deep UV scanners^[15]. Due to the high gas absorption of 13.5nm wavelength UV light, the EUV vessel has to be kept in high vacuum which may suffer the system with particle contamination in contrast with deep UV chamber filled by clean nitrogen. In addition, high energy laser beam will

evaporate liquid tin and produce tiny microparticles of tin vapor. These tin deposit contamination in the vacuum chamber was one of the biggest challenges in EUV system^[20]. ASML is implementing hydrogen buffer gas with pressure about 100 Pa around the collector mirror to decelerate tin ions and reduce atomic tin deposition by a chemical reaction: $\text{Sn}(\text{s}) + 4\text{H}(\text{g}) \rightarrow \text{SnH}_4(\text{g})$, these SnH_4 gas product can be pumped away from the chamber, as shown in Figure 4^[21].

ASML subsidiary Cymer and Japan-based Company Gigaphoton are long-term rivals, both are struggling to win the battle to develop high power, high conversion efficiency light sources for EUV lithography applications^[22].

The following paragraph describe main parts of a typical LPP system in a market-available ASML NXE 3x00 EUV system.

2.1. CO_2 Laser Generator and Power Amplifier

High power of input CO_2 infrared laser is crucial to EUV final power output, which gives a

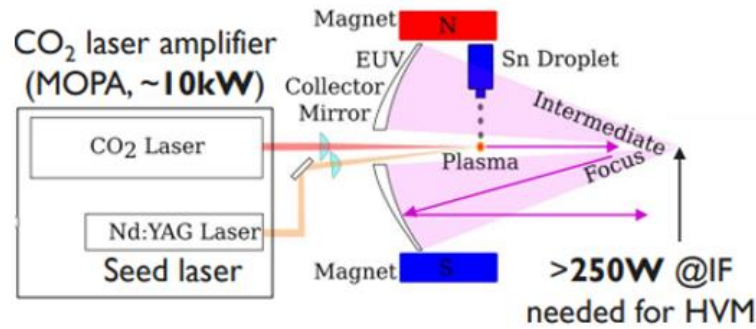


Figure 5. Illustration of EUV generator. Incident CO₂ and Nd:YAG laser beams hit tin droplet to generate dense plasma around the droplet, with the relaxation of plasma, EUV radiation takes place and focused by the collector mirror. Courtesy of Vu Luong, IMEC^[14].

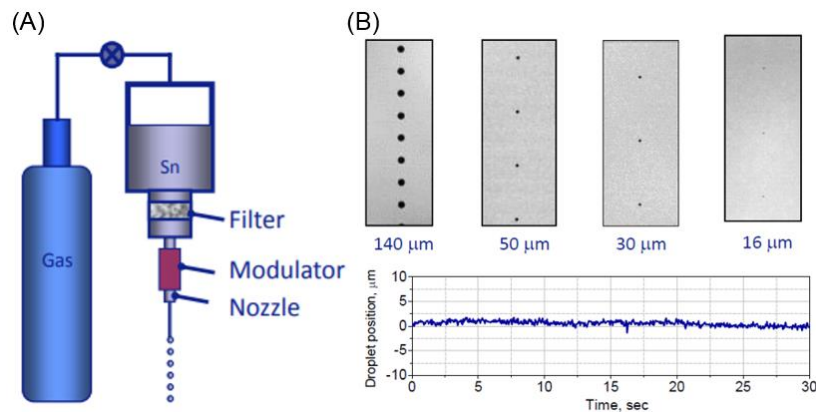


Figure 6. (A) Illustration of Sn droplet generator. (B) Snapshot of Sn droplets with different size and interval. Courtesy of Vu Luong, IMEC^[14].

second life of this mature laser technology^[18,19]. It was proven that tin plasma EUV could be driven by CO₂ laser pulse at 10.6 micron wavelength for the reasons of higher conversion efficiency, lower production of debris and higher average power levels without serious problems of beam distortions and nonlinear effects which occur normally in solid state lasers at high intensity. Practically a seed beam from Nd:YAG solid state laser at about 1 micron wavelength is preferred to be coupled together with the tin plasma as shown in Figure 4^[14]. An optimal timescale for an energy coupling was found to be a few nanoseconds for the Nd:YAG lasers and around ten nanoseconds for the CO₂ lasers.

The high power demand of a EUV source could be satisfied by a Master-Oscillator-Power-Amplifier system configuration. This new type of hybrid pulsed laser system combines pre-pulse seed laser, master oscillator, power amplifier (PA), fast flow systems, beam transport system and diffusion cooled planar waveguide lasers. The system can provide up to 40kW laser pulse, which could trigger 250W tin droplet EUV light source for latest ASML NXE

3400B EUV system with conversion efficiency (CE) up to 6%, so as to enable 140 wph (Wafer Per Hour) scanner throughput in HVM (High Volume Manufacturing)^[21]. New generation of 60kW modern is under development in labs.

2.2. Droplet Generator and Metrology

Inside the EUV source vessel, individual Sn droplets are continuously released with modulated size, speed and frequency from a nozzle of droplet generator (DG) located on top. As illustrated in Figure 5, tin raw material with high purity is melt in a container connected to the nozzle and pressed by an inert gas into the nozzle. Before spraying out of the nozzle, the tin jet is cut into droplets by mechanical vibration applied on in the nozzle under well-control. Separation space and size of droplets could be well-controlled by generator pressure and vibration frequency. Taking the nozzle outlet as a position reference, with the increasing distance from the outlet neighbor droplets coalesce together to from bigger droplets at larger separation space. The droplet size and timing must be precisely calculated

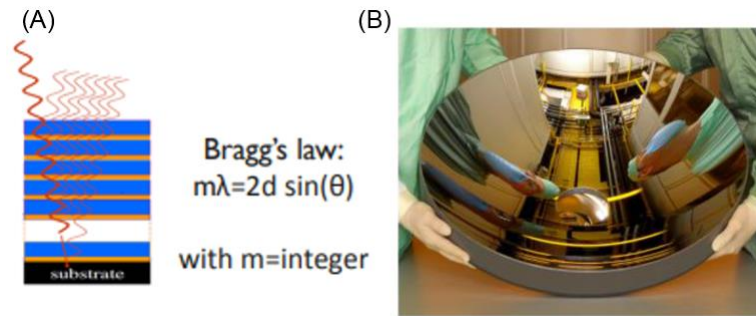


Figure 7. (A) Illustration of multilayer coated on collector mirror^[14]. (B) Photo of a real collector mirror from ASML^[21]. Courtesy of Igor Fomenkov from ASML and Vu Luong from IMEC.

and measured at the bombard position where laser pulse will hit the droplet to excite dense Sn^+ plasma. This angry-bird-like process play a critical role in EUV light source generation, and it has been found the EUV power is proportional to droplets speed and space between droplets. All components must be modulated with highest precision.

ASML third generation droplet generator produces with $27\mu\text{m}$ diameter of droplet with about 2700 hours average lifetime^[1].

2.3. Droplet Generator and Metrology

Plasma generated EUV light will be collected by an ellipsoidal MLM (Multi-Layer Mirror) collector with high reflectivity (Figure 7). The radius of the collector is well-designed to have its EUV focus on the vessel outlet called intermediate focus (IF) where EUV light is measured and conducted into the scanner vacuum chamber. In order to filter only 13.5nm EUV from the plasma halo and improve the reflective rate, collector surface is coated with multiple Mo/Si layers. By Bragg's law $n\lambda = 2d \sin(\theta)$, reflection wavelength λ could be precisely selected by adjusting layer thickness d and reflection angle θ . Main technical challenges include sputtering thickness and uniformity control of MLM layer, sputter target material purity, super fine polishing for surface roughness control, deposition of debris and contaminants, material decomposition and oxidation under EUV. The most advanced Bragg reflector of Mo/Si system can reach peak reflectance up to 70% (theoretically) of 13.5nm EUV light.

2.4. System Power Efficiency

EUV light source power is critical for high volume manufacture. To get an acceptable throughput of over 100 wph, EUV source above 200W @ IF (Intermediate Focus) is needed. Key factors for high source power are high input CO_2

laser power, high conversion efficiency (CE), high collection efficiency (reflectivity and lifetime) and advanced controls to minimize dose overhead, which follows below formula^[1,23]:

$$\text{EUV Power} = (\text{CO}_2 \text{ laser power} * \text{conversion efficiency [\%]} * \text{transmission}) * (1 - \text{dose overhead [\%]})$$

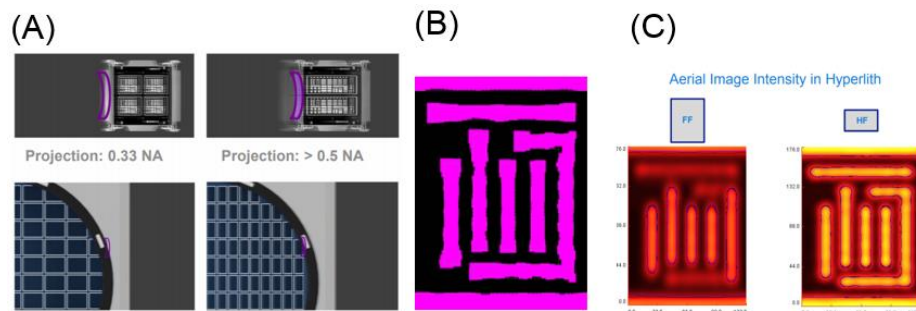
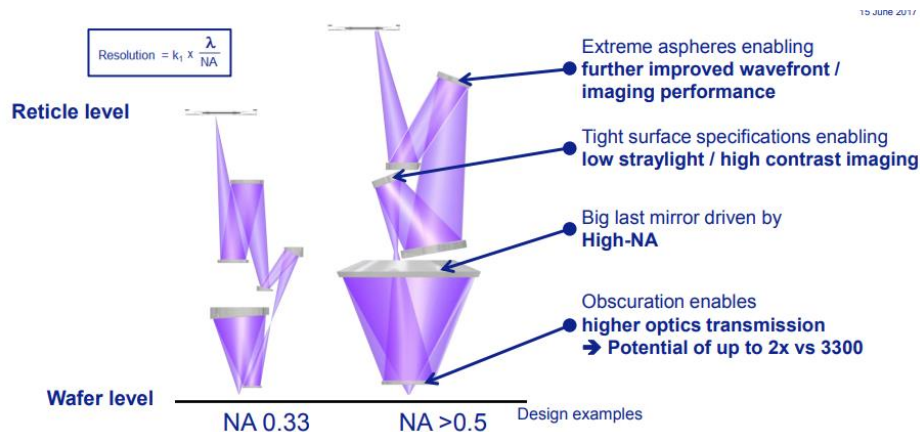
Latest ASML NXE3400B system adopts a 40kW CO_2 laser with system CE up to 6% and 10% dose overhead, which could generate 250W EUV light source. This enables wafer throughput up to 140 wph already acceptable for HVM. 400W EUV light source with 0.55 NA is under development at ASML, which will enable 185 wph throughput^[1].

3. Reflection System and High NA

Due to the strong absorption of 13.5nm EUV to any gases, the whole optical path must be fully reflective and enclosed in a vacuum environment, which is different from that of a deep UV scanner system. Moreover, in order to further reduce absorption and boost reflectivity, metal-based reflection mirrors is coated with alternating molybdenum and silicon (MoSi) layers with several nanometer in thickness (called Bragg mirrors), similar as the collector in EUV light source vessel. High reflectance is achieved with careful control of mirror quality, layer thicknesses, multilayer materials, interface quality, and surface termination. After several years of research and development, reflectance and film properties are relatively stable and can satisfy the requirements of an advanced EUV lithography system. The reproducibility of the reflectance peak was characterized to be as small as 0.2 percent^[24]. The coating uniformity has been improved to be better than 0.5 percent across a 150 mm diameter substrate^[25].



Figure 8. Internal structure of ASML NXE:3400B scanner. Source: ASML.

Figure 9. (A) Anamorphic half field (HF) exposure with high NA compared with full field (FF) with 0.33 NA. (C) shows aerial image intensity mapping between FF and HF A from a test mask feature (B), where HF presents much higher contrast^[28]. Courtesy of Jan van Schoot, ASML.Figure 10. EUV reflection comparison between NA=0.33 and high NA mirrors^[26]. Courtesy of Anthony Yen, ASML.

Reflection optics is also critical to next generation EUV scanner with high NA (> 0.5). High NA system will improve the wafer throughput to above 160 wph with dramatically reduced multi-pattern masks (number of LE), as shown in Figure 9. However, to reach high NA, reflection mirrors have to be redesigned to be asphere and the size of the mirror will be dramatically enlarged, accompany with EUV power over 400W^[26]. Besides, anamorphic half filed reticle and exposure system

has been introduced with 8x Y-magnification instead of traditional 4x magnification in both X and Y direction (Figure 9). Accordingly mask stage and wafer stage acceleration have to be adjusted to support twice number of scans/shots on wafer. Figure 10 illustrates mask comparison between 0.33 NA and 0.5 NA and wafer shot difference.

High NA EUV will benefit wafer throughput and reduce mask layers, but at a cost of super high EUV light power, very large mirror, asymmetry and

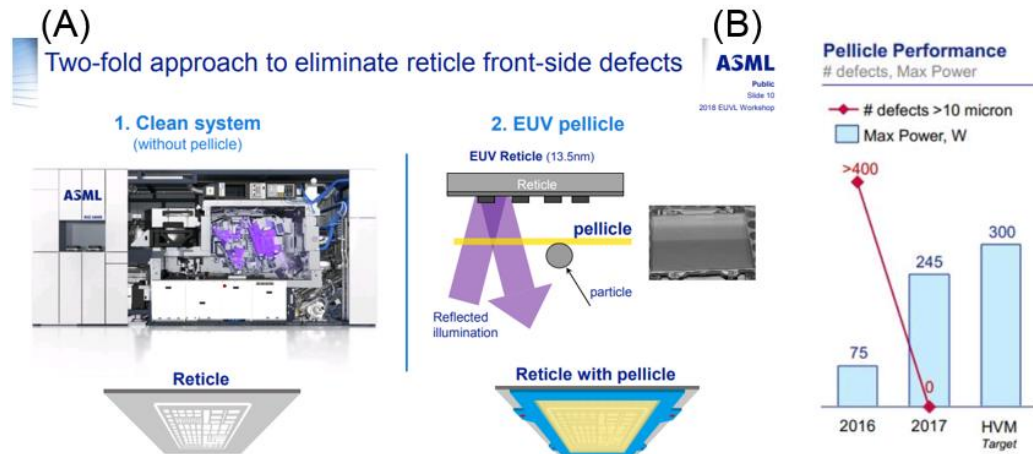


Figure 11. (A) An ASML presentation slide illustrates pellicle application on an EUV reticle and how particle can be isolated from reticle surface by a pellicle. (B) Defects decreased with the application of pellicle^[26]. Courtesy of Anthony Yen, ASML.

complicate reticle and wafer stage, as well as a special wafer cooling system. ASML future EXE module is supposed to be equipped with high NA optics^[26-28].

4. EUV Mask

In the EUVL reflective optical system, a reflective reticle is crucial to imaging quality and lithography process window on wafer. Similar as multiple coating on reflective mirrors in the vacuum system, 40 pairs of Si/Mo layers are deposited on the substrate of an EUV mask in order to get maximum reflection, which are capped by a thin film of Ruthenium (Ru) as thermal emission and reinforcing layer^[25,29]. On top of the thermal emission layer, absorber layers will be patterned to form “clear” or “dark” features. Material selection of absorber is critical to maximize absorption and minimize reflection. Shadowing effect and mask 3D effects are also related to absorber layer thickness and side-wall profile^[30]. Moreover, UV, thermal and mechanical stability of the reticle also need to be considered.

4.1. Pellicle Application

Usage of pellicle can prevent reticle from contamination and particles in the vacuum system, and greatly improve the lifetime of a reticle. But pellicle membranes on top of a reticle will reduce the EUV transmission and add extra cost to a reticle. ASML is struggling in both approaches: improve vacuum system cleanliness to avoid particle generation, and develop high performance pellicles membranes with high EUV transmission and low

defect^[26,28]. Progress has been made in recent years which makes the 1st generation pellicle ready for HVM which has been firmly tracked by Intel and other main EUV customers.

Advanced materials has been developed for pellicle membranes which should be durable under high power (> 300W) EUV radiation. SiNx is found to be a good backbone rigid materials for mechanical stability, Ru-capped SiN and Graphene sheet have been investigated at ASML and IBM, Graphite film is being developed at Samsung, Carbon Nanotube (CNTs) thin film composite with SiNx as a pellicle membrane has been developed and evaluated deeply at Hanyang University and IMEC. CNT enhanced SiNx membrane was found to have 2.5 times decrease of deflection when pressure applied, without scarify EUV transmission^[31].

4.2. Mask Substrate and Bragg Mirror Layers

Physical properties of quartz glass based backbone influence aerial image deformation, and resist line edge roughness (LER) as well. Substrate macro properties like bow, material stress, surface roughness and flatness is determined by manufacturing process, e.g. lapping, cleaning, and fine CMP polishing. According to analysis from Applied Material, around 75% blank defects are introduced during manufacturing process of the substrate, compare to 25% of that from the deposition of multiple layers^[25]. On the other hand, heat transmission and high temperature mechanical stability of materials are also concerned. Applied Materials has taken a lot of efforts on reticle blank material investigation and development and already

Current EUV Mask Blank Structure

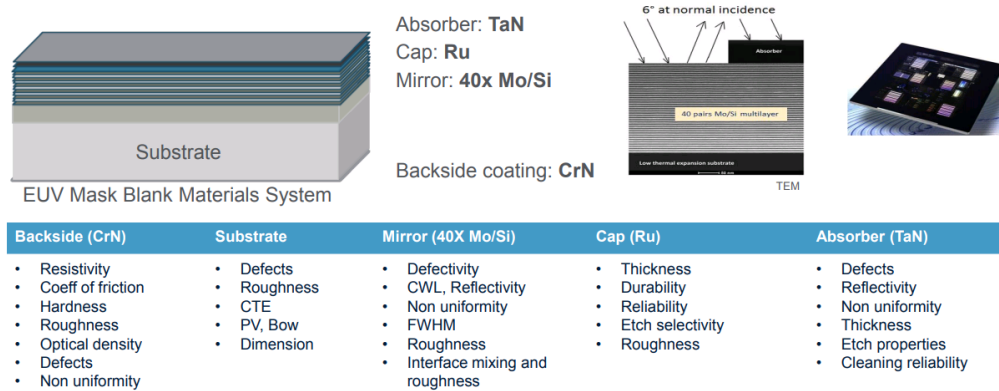


Figure 12. EUV reticle structure illustration from Applied Materials^[29]. Courtesy of Vibhu Jindal, Applied Materials.

made great progresses to improve substrate flatness, surface roughness, bow and defectivity to HVM level during last two years^[25,29].

Si/Mo switching Multilayers are sputtered on top of the substrate, which generate stress within multilayers and substrate interface as shown in Figure 12. One method is to selectively cut continuous mirror layer on uncritical area to release stress. Other critical properties which need to be preciously controlled are layer uniformity, thickness, interfacial roughness and defects within multilayers. It was observed that silicide formed within Si-Mo interface which could deteriorate EUV reflectivity, an optimized deposition process has been developed by Applied Materials to limit the silicide within sub-nm region, and such maximize reflectivity of EUV light^[32].

A capping layer on top of mirror layer is needed to enhance mechanical durability and reliability of the reticle, and improve the adhesion between mirror layers and the absorber. Ruthenium was implemented as a capping layer in state-of-the-art EUV reticles, other advanced materials or composites have been investigated as well within major plays like Applied Materials and Veeco^[33].

4.3. Absorber and Mask 3D Effect

Mask 3D effect in EUVL is more predominant for wafer image quality, compared with that in DUV lithography. Thickness of absorber and Chief-ray Angle (CRA) introduce asymmetry EUV reflection, mask shadowing and different printing behavior between horizontal and vertical features, as shown in Figure 13 where horizontal patterns A suffers much higher shadowing effect than vertical pattern B due to off-axis incidence angle only in Y direction. Mask

3D effect is also a result of multilayer mirror reflectance changes over incidence and wavefront or phase deformation of the material. In current EUV scanner modules, horizontal and vertical lines show different CD variation and contrast^[14].

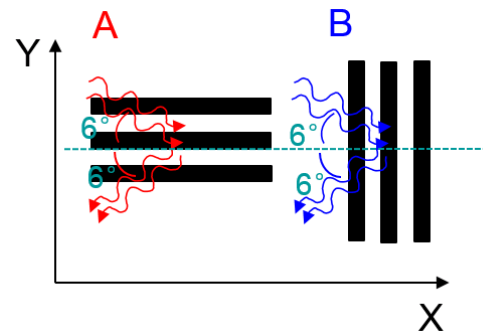


Figure 13. Feature orientation dependent mask 3D effect, horizontal features suffer more since they are perpendicular to the incident light plane.

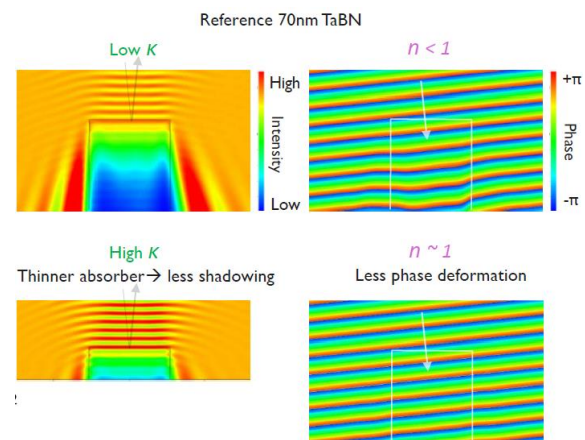


Figure 14. High k absorber can reduced mask 3D effect and get less phase deformation compared with low k absorber^[14]. Courtesy of Vu Luong, IMEC.

Ideal EUV absorber material should have a high extinction coefficient (k) and extremely low reflectivity, as well as small phase deformation after longtime EUV radiation at the same time, so that a reduced thickness and diminished 3D effect could be obtained. Boron doped TaN with about 70nm thickness is currently adopted for HVM EUV application with satisfactory results. But alternative absorber materials is also under evaluation. Today high- k absorbers such as Ni, Al, Co and their alloys have been evaluated by Veeco, IMEC^[34], ASML together with mask shops like Toppan and Hoya^[35], amorphous single phase materials as absorber is under development by Applied Materials^[29], which shows low surface roughness and stress, good chemical durability, good adhesion and etch selectivity with capping layer and less than 2% reflectivity for less than 45nm thickness. Pt-CrN Multilayer absorbers used for phase-shift mask (PSM) has been investigated by Hanyang University^[36]. On the other hand, EDA vendor Synopsys has evaluated absorber side-wall angle impact on optical parameters like NILS and intensity latitude by simulation^[30].

4.4. Actinic Inspection and Defect of An EUV Mask

Mask defect is one of the major challenges limiting the high volume manufacture of EUVL system. Since EUV light will penetrate into multilayer structure on the mask and reflected back, both internal and surface defects have to be taken into account of amplitude and phase shift of the reflected EUV light. Each component of a mask multilayer can contribute to final defect formation. As mentioned in previous paragraph, the quality of a mask substrate is decided by its bow, surface flatness and roughness. Impurities or surface roughness can generate dislocations and lattice boundary on the substrate surface which can be easily extended to upper layers of the reticle and eventually reticle surface. Uniformity of the multilayer Bragg mirror may suffer from pits or bumps buried within 80x multilayers during sputtering deposition due to any contamination inside the chamber, gas or target impurities, thickness also vary on tool variation, so as the absorber, capping layers and pellicles. Contrary to deep UV masks, the surface roughness of an EUV mask causes phase variations of the incoming wave fronts, which lead to intensity variations in the aerial image i.e. speckles which are local intensity inhomogeneities^[37,38]. Because of the

reflective nature and much smaller wavelength, the speckles contrast is significantly higher than that of the deep UV masks and may increase Line Width Roughness (LWR) at wafer level.

Fortunately not all defects or impurities in multilayers (ML) will result in disaster of reticle scrap, some substrate defects can be de-magnified when they spread onto the reticle surface through ML of the mirror^[39]. Some defects from multilayers can be locally repaired by E-beam similar as that for deep UV masks, as illustrated in Figure 15.

On the other hand, EUV mask characterization and actinic review of defects is also critical for HVM of EUVL since traditional SEM direct characterization cannot detect light phase changes and intensity variation due to mask internal defects or surface roughness. EUVL infrastructure consortium among ZEISS, SUNY Polytechnic Institute and SEMATECH has developed an actinic EUV mask Aerial Image Monitor System called AIMS™ EUV which is trying to close the gap^[37]. With the help of AIMS system, aerial image of the mask could be obtained before wafer exposure, any phase changes or surface roughness from will be captured and their impact on wafer and line edge roughness will be evaluated automatically.

5. EUV Photo Resist

Stochastic effect of photo resist is another major concern for EUVL application^[40-42]. Due to limited light source power, short wavelength and high photon energy of EUV light, Auger electrons or 2nd electrons will be generated after photon absorption by the resist, resulting in uncontrolled free electron path and photo acid spread after electronic relaxation, which is considered to be the main root cause of line edge roughness and CD variation^[43,44]. Interdisciplinary teams worldwide have been working on understanding fundamental mechanism of photo chemical reaction, new material development, and process control.

Gupta team from UCLA focus on LER analysis and impact on FEoL and BEoL of advanced FinFET technologies, LER mathematic model and LER impact on device reliability have been investigated^[45]. Resist half-pitch (HP) resolution, dose sensitivity and their relationship (Figure 16) to LER have been evaluated by Paul Scherrer Institut (PSI) to get a comprehensive solution^[46]. It was found that amount of photons, quencher and acid rank top three LER contributor, LER could be

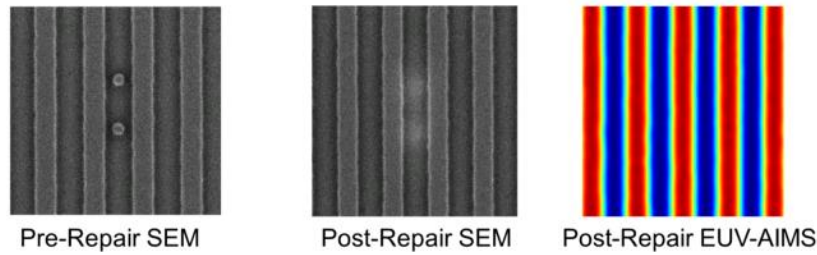


Figure 15. Defect on EUV mask could be repaired. From left to right, it shows two dot defects on line-space features, defects removed, and an aerial image after reparation^[37]. Courtesy of Dirk Hellweg, Zeiss SMT.

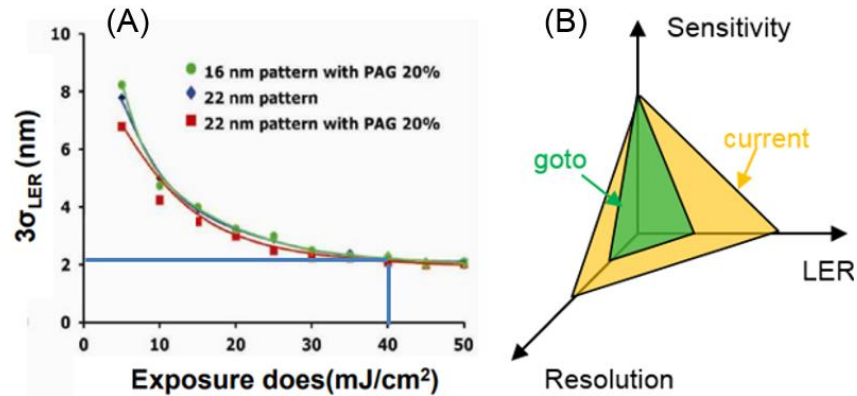


Figure 16. (A) LER is strongly dependent on exposure dose. (B) Resist sensitivity, resolution and line edge roughness (LER) are correlated each other, an optimization of smaller LER with good resolution and sensitivity is needed for EUVL.

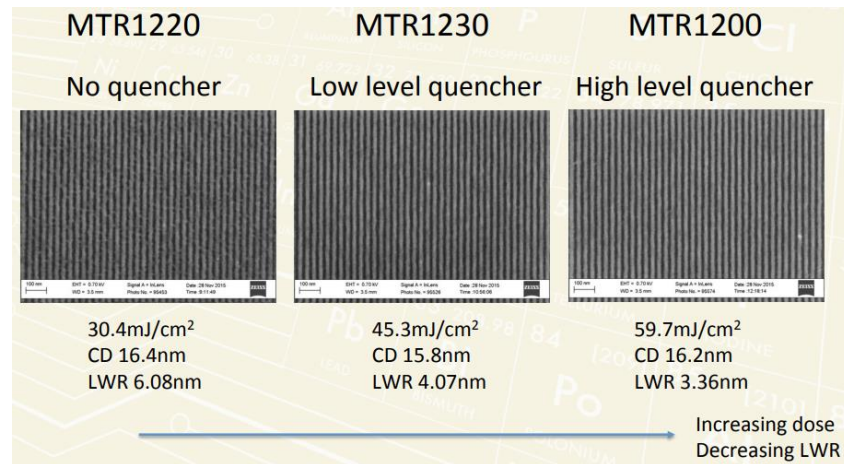


Figure 17. CDSEM images show LER is decreasing with increasing dose and resist quencher concentration^[48]. Courtesy of C. Popescu, University of Birmingham.

reduced by optimizing multi-trigger component ratio and quencher amount of the resist chemicals^[47,48].

Metal-contained photoresist, chemical amplified resists (CARs), photo acids (PAs) together with their backbone polymers have been investigated, material quantum yield and absorption coefficient have been measured and correlated to electron mean

free path within resist, providing a theoretical guide for photoresist selection. Resist chemical processes after EUV radiation are widely investigated by teams from Berkeley lab, Molecular Foundry, LBNL, Columbia Hill Technical Consulting and IBM Almaden Research Center^[49,50]. Numeric model of imaging mechanism for sol-gel prepared

organometallic polymer resists have been created by Inpria Corp., stochastic diffusion of secondary electrons has been simulated with changing dose^[51]. Innovative Inorganic-organic composite photoresist with a metal oxide core and polymer ligands has been synthesized as a non-CAR resist which is supposed to have smaller LWR ($3\sigma < 3\text{nm}$)^[52-54]. Moreover, lithography process control and optimization are also essential to LER reduction. Toshiro Itani and Takahiro Kozawa from Japan have conducted experiments to improve line edge roughness by alternating development, bake, rinse solution as well as resist top and bottom coating^[55].

Outgassing is another issue during resist exposure, hydrogen-contained molecules can contaminate the vacuum chamber, mirrors and reticles. ASML has developed a Dynamic Gas Lock (DGL) membrane to prevent outgassing spread and suppress DUV and IR during the wafer exposure^[1].

6. Summary and Outlook

In this paper, we had a broad but not specialized overview of extreme ultraviolet lithography technology and its application in mainstream semiconductor wafer manufacturing. Major EUVL modules include light source and vessel, reflection mirror system under vacuum, reflective reticle and aerial inspection system. Multilayer material fabrication and photoresist characterization are also critical in terms of EUV patterning performance, defectivity and edge roughness control. As EUVL scanner is an integrated system, final wafer printing quality is determined by every component especially the weakest ones. Thanks to worldwide collaboration and big investments in recent years, critical issues related to light source, substrate material and process, multilayer defects, mask absorber and defects, as well as resist edge roughness have been overcome in an ASML EUV scanner system, and wafer HVM using these EUV tools are already realized in those leading fabs worldwide.

Major progresses and challenges that are related to wafer throughput and pattern printing quality are summarized below.

EUV light source power

250W light power boosts wafer throughput to 140wph, which makes ASML EUV scanner ready for HVM. Next generation EUV source with further increased EUV energy power, improved conversion efficiency and better dose control is under development, not only to raise wafer throughput but

also to further minimize LER which is closely related to dose on wafer. It was found $40\text{mJ}/\text{cm}^2$ dose (compared with $20\text{mJ}/\text{cm}^2$ from current module) can dramatically reduce LER to smaller than 3nm. 400W EUV source will be implemented in next module of ASML NXE scanner soon. High power EUV is also critical to next generation high NA scanner.

Mirror and reticle defectivity

Impurities, thickness inhomogeneity of multilayer mirrors and reticles can change EUV light intensity or phase which will generate defects on wafer. Reticle surface roughness also contributes to LER. Besides superb quality fabrication of multilayer mirrors and reticles, the control of contamination and particles in vacuum chamber is also a big concern since there is no clean pressure gap protection in the reflection chamber and high energy nature of EUV. With the introduction of pellicles and DGL membrane, chamber defects have been reduced to a very low level. In order to detect not only surface impurities but also light phase changes due to internal defects, AIMS inspection system has been developed by ZEISS together with SEMATECH. This tool could check wafer aerial image and detect potential pattern failure on wafer.

LER and CD variation

Compared with deep UV lithography, line edge roughness is much worse for EUVL due to relatively low dose and secondary electron diffusion (stochastic effect). New photo resists such as metal-oxide with polymer ligands are being developed, lithography process flow has been optimized, and reticle flatness is well-controlled, all efforts have successfully reduced the LER to an acceptable level for an ASML EUV system. Further investigation is trying to reduce LER to below 3nm for 3σ from current 3.5nm range from inorganic resist and 5nm range from CAR.

Mask 3D effect

Due to absorber thickness and the chief-ray angle of incident EUV light onto the mask, horizontal and vertical features show different imaging behavior and critical dimension change on wafer, this is called shadowing effect. Mask 3D is also related to reticle blank flatness and bow, multilayer EUV reflectivity, and absorber properties. These effects will decrease the margin of process window with a reduced or shifted depth of focus (DOF). By introducing phase control SMO and OPC

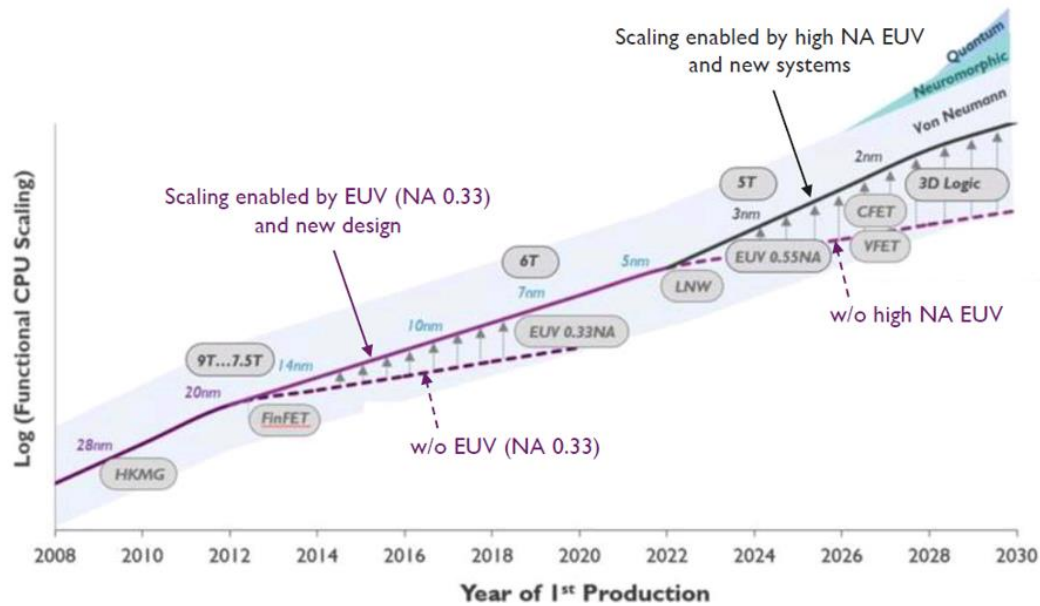


Figure 18. Technology node scaling is driven by the development of lithography and DTCO. High NA EUV will lead the industry in next decade^[14]. Courtesy of Vu Luong, IMEC.

correction, orientation-dependent mask CD bias could be compensated. High-k absorbers are also under development to help further reduce shadowing effect. In next generation of high-NA EUV scanner, anamorphic half fielded reticle will be introduced to overcome x-y asymmetry and improve throughput^[56], but with the cost of design, OPC and mask complexity.

EUVL has achieved great progress since 2015, most of breakthroughs take place after the formation of EUV industry alliances among big player like ASML, INTEL, TSMC and SUMSANG. Industry requirements combined with strong financial support finally drive this technology into HVM. As the only EUVL scanner commercial provider, ASML is going to sell over 30 EUV modules in 2019. The market is ramping up and this revolutionary technology will lead the industry to the next decade^[57] (Figure 18).

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Chapter 1

Introduction

1.1 Historical Background

For several decades, the resolution capability of optical lithography was extended by a succession of transitions to shorter imaging wavelengths. The first commercially available wafer stepper operated at a visible wavelength (436 nm), while later generations of exposure tools imaged at mid-ultraviolet and deep-ultraviolet wavelengths (Table 1.1). During efforts to continue this process of improving resolution by using a wavelength of 157 nm, with F₂ excimer lasers as the light sources, a number of difficulties were encountered, and optical lithography was eventually extended not by using light with a wavelength of 157 nm but by adapting ArF lithography to an immersion configuration that enabled the numerical aperture (NA) to be increased to 1.35. There were some additional efforts to increase the numerical aperture of ArF immersion lithography further, but development activities were eventually suspended because progress was slow.¹ The shortest wavelength used in optical lithography continues to be 193 nm.

In spite of the lack of success in developing HVM-capable lithographic technologies at 157 nm (or shorter optical wavelengths), lithographers were not ready to discontinue the pursuit of extending lithographic capability. Considerable amounts of money and engineering effort were invested in developing x-ray lithography, with a very short wavelength of ~1 nm.

Table 1.1 The wavelengths used for lithography.

Wavelength	Light source	Year of first use in high-volume manufacturing (HVM)
436 nm	Mercury arc lamp (g-line)	1982
365 nm	Mercury arc lamp (i-line)	1990
248 nm	KrF excimer laser	1994
193 nm	ArF excimer laser	2001
13.5 nm	Laser-produced plasma	2019

Unfortunately, projection optics at this wavelength could not be fabricated, principally because there are no materials that are sufficiently transparent at $\lambda \approx 1$ nm to form refractive optics, or have sufficient reflectivity for near-normal incident light, thus precluding reflective optics. This meant that x-ray lithography involved proximity printing, resulting in two significant problems. First, the patterns on wafers are replicated 1:1 from the mask, so mask quality, in terms of registration, dimensional control and defects, had to be exceedingly good. Second, fundamentally and ultimately more consequential, were the diffraction effects that became insurmountable for printing features with dimensions < 100 nm.

These problems with extending optical lithography to wavelengths shorter than 193 nm involved the lack of suitable transmissive or reflective optical materials for lenses, masks, and pellicles for enabling projection lithography with reduction lenses. However, in the 1980s, multilayer films were developed that provided practical reflectivities at wavelengths $4 \text{ nm} < \lambda < 25 \text{ nm}$.^{2,3} This development led to proposals for lenses with all-reflecting optics that could be used for projection lithography.^{4,5,6} Initially, this was referred to as soft x-ray lithography, but the name “extreme ultraviolet” was later adopted to distinguish this new type of projection lithography from proximity $1 \times$ x-ray lithography. A nice review of the early history of EUV lithography was written by two pioneers of this technology, Dr. Obert Wood and Dr. Hiroo Kinoshita.⁷ Another set of key early activities involved a consortium, the EUV Limited Liability Corporation (LLC), which produced the first full-field EUV exposure system.⁸

The consortium Sematech played a critical role in the next phase of EUV lithography development. In addition to co-sponsoring annual EUV lithography symposia, Sematech engaged in development activities directly. A small pilot line was established to produce low-defect EUV mask blanks.^{9,10,11,12} The associated metrology capabilities were used to support mask blank fabrication at commercial suppliers. Sematech also funded an EUV Micro-Exposure Tool (MET) that was placed at a synchrotron light source at Lawrence Berkeley National Laboratory.^{13,14} The EUV MET provided essential capability for improving EUV resists.^{15,16}

Another key event in the development of EUV lithography was the decision by ASML to build a pair of full-field exposure tools, the Alpha Demo Tools (ADTs). These were installed at the consortia INVENT in Albany, NY and Imec in Leuven, Belgium, further expanding access to EUV exposure capability. The full-field exposure capability enabled the accomplishment of an important milestone in moving EUV lithography from the laboratory to the factory: the fabrication of functioning SRAMs on a test chip from Advanced Micro Devices with an interconnect layer patterned with EUV lithography.¹⁷

Chapter 2

Sources of EUV Light

In Chapter 1, it was explained that the wavelength of light at which EUV lithography is practiced was chosen largely by the availability of multilayers with high reflectance. This can be contrasted with the situation in optical lithography, which has always been operated at wavelengths where there are strong sources of light with narrow bandwidths. The result is that EUV lithography has suffered from light sources that are relatively weak, and a great deal of R&D has therefore gone into the development of EUV light sources in order to improve the situation.

Several methods have been used to generate EUV light. Early researchers used synchrotron light sources, but smaller light sources became needed as a practical matter as EUV lithography moved from research to development. This was true for metrology applications as well as for exposure tools, and stand-alone systems provided greater flexibility for development pilot lines. Moreover, it was eventually determined that synchrotrons of a practical size could not provide sufficient EUV light to be cost-effective, even with the use of undulators or wigglers.¹ The EUV source used currently in manufacturing tools, the laser-produced plasma (LPP) source, is discussed in this chapter, and a source that might be used in the future for wafer exposures, the free-electron laser (FEL), will also be described. Plasma-based sources used in a number of EUV metrology tools will also be described.

2.1 Laser-Produced Plasma Light Sources

The concept behind using plasmas to generate EUV light is that energetic photons can be emitted when electrons combine with highly (positively) ionized atoms. In laser-produced plasma light sources, ions are created by the strong electric fields of intense pulses of coherent laser light. These rapidly oscillating electric fields cause repeated collisions between electrons and ions, creating ions stripped of multiple electrons (see Problem 2.1). Energetic photons are produced when electrons recombine with these highly charged ions. Although atomic emissions typically occur at quantized wavelengths, for

plasma-produced light, the emissions can occur over a range of wavelengths because the kinetic energy of the recombining electrons and ions also contributes to the energy of the emitted photons. The net result is emissions over a broad range of wavelengths. This is fortunate because an exact match between high multilayer reflectivity and a sharp emission line would be difficult. On the other hand, this also means that much of the light produced is not useable for wafer exposures, limiting the efficiency of plasma sources. Moreover, plasmas will produce long-wavelength DUV light that can expose resist if transmitted to the wafer plane.

The target material, often referred to as the fuel, is chosen by considering several factors. Conversion efficiency from laser light to in-band EUV light is one of the most important considerations. Xenon, tin, and lithium plasmas were found to have strong emissions in the EUV portion of the electromagnetic spectrum. A xenon jet was used in the source of the first full-field EUV exposure tool, the Engineering Test Stand (ETS), that was produced by the EUV LLC. Xenon was thought also to have an advantage in terms of optics contamination, because xenon is a noble gas. Protecting optics from chemical contamination is needed because there are no glass windows that can be used to separate the source and optics. In practice, considerable erosion of collector mirrors was observed in light sources with Xe as a fuel, even though a noble gas was used. Experiments revealed that the erosion was caused by sputtering by Xe ions with high kinetic energy.²

Because the mirrors needed protection from even chemically inert materials, measures were developed to reduce significantly the amount of material from the plasma (often referred to as “debris”) that reaches the condenser mirror. Consequently, non-inert materials could be considered as fuel. This led to increased consideration of tin as a fuel, because it has stronger EUV emissions than xenon, and EUV light is produced by multiple charge states of tin ions, while appreciable emission near $\lambda = 13.5$ nm is produced by only a single Xe ion (see Figs. 2.1 and 2.2). Eventually, tin became the material used in all sources for ASML’s 0.33-NA exposure tools, although Xe continues to be used for some EUV light sources, particularly for metrology (see Section 2.2).

The Engineering Test Stand (ETS) that was produced by the EUV LLC featured an LPP light source that used a pulsed Nd:YAG laser (1064-nm wavelength) to generate the plasma. There were significant doubts regarding the possibility of scaling up the power using this type of laser, and the operating and maintenance costs were high. This led to questioning LPP sources in general and resulted in substantial efforts to develop discharge-produced plasma (DPP) EUV light sources. However, these systems were unable to meet light output and reliability targets for exposure systems, so LPP sources were reconsidered. An example of a DPP source will be discussed in the next section.

Chapter 3

EUV Exposure Systems

EUV lithography exposure systems are conceptually similar to optical ones, involving light sources (discussed in the previous chapter), illumination and projection optics, alignment and focus systems, wafer stages, and sub-systems to mechanically handle wafers and reticles. Most of these need to be designed to meet the specific requirements for operation at EUV wavelengths, and the focus of this chapter is on those aspects specific to EUV lithography.

Currently, only one exposure tool supplier, ASM Lithography (ASML), is producing EUV exposure systems for use in high-volume manufacturing (HVM), and all of the projection optics are provided by Carl Zeiss. Consequently, many of the details presented in this chapter regarding EUV exposure systems are the result of specific choices made by ASML or Zeiss, while at other times, system characteristics are compelled by the physics of EUV lithography, independent of supplier. Throughout the chapter, it will be noted what is required generally and what is used specifically in the systems built by ASML. For example, the systems currently available from ASML have $NA = 0.33$, while systems with $NA = 0.55$ are under development. For both current and future systems, there was latitude for choosing the NA, but actual exposure tools have specific numerical apertures due to the choices made by the equipment and optics suppliers.

3.1 Lithography in Vacuum

An important and essential characteristic of EUV lithography systems is that exposures take place with wafers, lenses and masks under vacuum. EUV light is strongly absorbed by air, so there must be vacuum, or very low pressures, wherever actinic light is going to transit in EUV exposure tools. (The transmission of 13.5-nm light through 1 mm of air at atmospheric pressure is only $\sim 0.1\%$.) There are a number of ramifications of vacuum. In optical exposure tools, air and gases perform a number of tasks, including thermal, chemical contamination, and defect control, and different approaches are required for addressing these issues in EUV exposure systems.

Temperature control is essential for stable exposure tool operation. In optical exposure tools, much attention has been paid to lens, wafer, and mask heating, since lens heating can lead to aberration errors, particularly defocus, as well as overlay errors, and the heating of wafers and masks also has an impact on overlay. Air flow is one method by which lens and wafer temperature is controlled in optical exposure tools, but the vacuum requirements in EUV systems limit the applicability of this approach. Temperature control in vacuum is more difficult than in air.

Consider, for example, a flat, solid surface over which air is flowing, with a temperature difference ΔT between the air and the solid. The rate of heat transfer per unit time q is given by¹

$$q = h_C A \Delta T, \quad (3.1)$$

where h_C is the convective heat coefficient, and A is the area of the surface over which heat is being transferred. A useful approximation for h_C is given by

$$h_C = 10.45 - \nu + 10\sqrt{\nu}, \quad (3.2)$$

where ν (m/s) is the speed of the air relative to the surface, and h_C is measured in units of W/m²K. If we assume that there is a temperature difference of $\Delta T = 0.1$ K and $\nu = 1$ m/s, then the heat transfer per unit area is given by

$$\frac{q}{A} = 1.95 \text{ W/m}^2. \quad (3.3)$$

In EUV lithography systems, there cannot be air flows at pressures that are an appreciable fraction of atmospheric pressure. Gas flows at low pressure can be used to achieve levels of heat transfer that are sometimes useful, but the efficacy of such flows is much reduced compared to what occurs at atmospheric pressure. The remaining mechanisms of heat transfer are conduction and radiation. The rate of radiative transfer per unit area between two flat, parallel surfaces is given by²

$$\frac{q}{A} = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}, \quad (3.4)$$

where σ is the Stefan–Boltzmann constant, and ϵ_1 and ϵ_2 are the emissivities of the two surfaces, which are at temperatures T_1 and T_2 . If we assume $\epsilon_1 = \epsilon_2 = 1$, and that one surface is at a temperature of 300.0 K while the other has a temperature of 300.1 K, then the rate of heat transfer by radiation is approximately 0.2 W/m². Comparing this with Eq. (3.3), one can see that convection is a much more efficient means for controlling temperatures by approximately an order of magnitude.

Chapter 4

EUV Masks

As discussed previously, there are no high-transmission optical materials at EUV wavelengths due to limitations imposed by atomic absorption. As a consequence, all optical elements in the lens must be reflective for EUV lithography.¹ Similarly, EUV masks must be either reflective or stencil-type. The semiconductor industry has made reflective masks standard, as they were seen as more practical than stencil masks. This means that masks for EUV lithography are fundamentally different from those used for optical lithography. An extensive amount of engineering effort was directed at determining what the format should be for reflective EUV masks, and additional effort was required to make them suitable for use in high-volume manufacturing.

4.1 Structure of EUV Masks

A typical configuration for an EUV mask blank is shown in Fig. 4.1. A multilayer reflector is deposited on the front side of a flat, glass substrate. A capping layer, typically ruthenium, is deposited on the multilayer, and then the absorber is deposited on the capping layer. Because the absorber is typically metallic, optical contrast is low between the absorber and multilayer, making defect inspection difficult. Consequently, a thin film is often deposited on top of the absorber to increase the contrast at DUV wavelengths. An electrically conductive backside film is also deposited to enable the mask to be secured on electrostatic chucks in exposure tools. Electron micrographs of an early EUV reflection mask are shown in Fig. 4.2.

As in optical binary masks, EUV binary masks are divided into “dark” and “light” areas, but for EUV lithography, the light areas are reflective rather than transmissive, while the dark areas are covered by absorbers to prevent reflection. For early EUV masks, gold was often used as an absorber, as also used for many $1\times$ x-ray masks, even though lift-off processes were required for fabrication, due to the absence of a process for dry etching gold. Potential absorber materials that could be dry-etched were considered, such as chromium, which was a standard absorber material for optical masks, and TiN.

Chapter 5

EUV Resists

To date, most resists used for EUV lithography are chemically amplified resists, based on KrF and ArF resist platforms (although new concepts are also being given serious consideration). However, the demands for EUV resists differ from those of photoresists used at optical wavelengths in some critical ways. Because EUV lithography is intended to be used to produce features and pitches beyond the resolution limits of ArF lithography, EUV resists must be capable of very high resolution and have low line-edge roughness. There is also an enhanced need for sensitivity because of the low output of EUV light sources compared to excimer lasers, a problem described in Chapter 2. For DUV resists, sensitivity has been obtained through chemical amplification, and the greater the extent to which photoacids can diffuse and produce deprotection, the fewer photons are required to expose the resist. However, substantial diffusion will effectively blur the image, reducing resolution. To maintain a high-resolution image after post-exposure bake, there is a rule of thumb that the diffusion length of the photoacids in chemically amplified resists should be less than $\sim 16\%$ of the pitch.¹ Using this estimate as a guide, for 20-nm half-pitch technology, the diffusion length will need to be less than ~ 6 nm and must be even smaller for later nodes. In addition to having much tighter requirements for resolution and LER, the mechanisms of the radiation chemistry of EUV resists are different from those of optical resists, and these mechanisms for EUV resists are discussed in the first section of this chapter.

5.1 Exposure Mechanisms of Chemically Amplified EUV Resists

In KrF and ArF chemically amplified resists, incident photons are absorbed by photoacid generators, with acids produced as a direct consequence of this photoabsorption.² The radiation chemistry in EUV resists involves a somewhat different mechanism. EUV photons are highly energetic (~ 92 eV), so photoabsorption typically results in the generation of a photoelectron. Acids are produced as the result of collisions between electrons and

Chapter 6

Computational Lithography for EUV

Early work on EUV lithography primarily involved patterning at large values of k_1 , at least in comparison to what had been routine for optical lithography for many years. For example, EUV lithography was introduced into high-volume manufacturing at the 7-nm node, with ~ 20 -nm half-pitch, giving $k_1 = 0.49$, while k_1 has been considerably less than 0.4 for logic using optical lithography since the 32-nm node. Although there were some early studies indicating that an accurate description of imaging for EUV lithography was necessarily complex, most of the complications were seen to occur at low values of k_1 , so the computational aspects of EUV lithography required to address many issues received much less attention than other aspects of the technology for a number of years. However, in order to meet the requirements of the 5-nm node and beyond, many computational aspects of EUV lithography need to be considered, and there has been much recent work.

There are several reasons why computational lithography for advanced EUV lithography is more complex than it is for optical lithography. To begin, OPC for EUV lithography still has all of the requirements of conventional, optical OPC, such as the need to ensure good targeting of dimensions at the best operating point and good dimensional control throughout the process window, which includes variations on the mask. Because EUV lithography is applied at the most advanced nodes, accuracy requirements are necessarily tighter than they are for optical lithography. There are some differences between the conventional OPC for EUV lithography and that for optical lithography. For example, it is important for both lithographic technologies to ensure that the image log-slope maintains a certain minimum, but this metric may get greater emphasis in EUV lithography. As noted in Chapter 5, the impact of resist stochastics can be reduced by maintaining a large normalized image slope, so more weight may be given to this image metric during OPC optimization.

Chapter 7

Process Control for EUV Lithography

There are several sources of process variation that are unique to EUV lithography or of significantly greater significance than in optical lithography. The issues for overlay, linewidth control, and defects relevant to EUV lithography are discussed in this chapter. Recently, attention has been given to the subject of edge placement errors, which result from a combination of critical dimension and overlay errors.¹ Although edge placement is the fundamental quantity of interest, overlay and critical dimension control will be considered separately in this chapter so that the causes of process variations can be understood more clearly. Metrology is an important consideration for process control and is the topic of the next chapter.

The application of EUV lithography only to very advanced nodes (7 nm and beyond) means that process control requirements are always very tight. Selected dimensions and control requirements from the 2020 International Roadmap for Devices and Systems (IRDS) are shown in Table 7.1. As can be seen, control is expected to be required at the nanometer and sub-nanometer

Table 7.1 Values for selected critical dimensions and control requirements for logic from the 2020 International Roadmap for Devices and Systems (IRDS).

Logic industry “Node” labeling (nm)	“7”	“5”	“3”	“2.1”	“1.5”	“1.0”
Year of production	2018	2022	2022	2025	2028	2031
MPU/ASIC minimum metal half-pitch (nm)	18.0	15.0	12.0	10.5	8.0	8.0
Metal LWR	2.7	2.3	1.8	1.5	1.2	1.2
Metal CD control (3 σ) (nm)	2.7	2.3	1.8	1.5	1.2	1.2
Gate LER	0.8	0.7	0.6	0.5	0.5	0.5
Physical gate length for HP Logic (nm)	20	18	16	14	12	12
Gate CD control (3 σ) (nm)	1.1	1.0	0.9	0.7	0.6	0.6
Overlay (mean + 3 σ)	3.6	3.0	2.4	2.1	1.6	1.6

level, that is, at the molecular scale (see Chapter 5). Meeting these requirements is challenging, particularly because there are mechanisms that induce process variations for which there are no counterparts in optical lithography, as will be seen in this chapter. Improvements of just a few tenths of a nanometer are significant for advanced nodes. For example, reducing the overlay by only 0.5 nm at the 5-nm node is a 17% improvement relative to the target.

7.1 Overlay

For overlay, all of the issues that are relevant for optical lithography pertain to EUV lithography. These include matters such as alignment and overlay measurement mark quality, overlay modeling, and mask and wafer heating. Some contributions to overlay errors, such as aberrations, are relevant for optical lithography but are more significant for EUV lithography. There are also some additional contributions to overlay errors for EUV lithography for which there are no counterparts in optical lithography, and these will be discussed in this section.

As noted in Chapter 4, mask nonflatness causes overlay errors in EUV lithography. For optical photomasks, an overall flatness of 100 nm is a typical requirement for leading-edge masks, driven by focus control considerations, with tighter flatness adding to blank costs. For EUV masks, much tighter flatness is required because of the impact on overlay. For example, only 50-nm nonflatness will lead to 1.3-nm overlay error at the wafer level (see Problem 4.3). This is a substantial fraction of the total overlay budget at the 7-nm node and beyond (Table 7.1). Without some method of compensation, for mask nonflatness to contribute an overlay error that is only 10% of the total overlay budget for the 5-nm node, mask flatness would need to be <10 nm. This would lead to extremely expensive masks, particularly since the additional polishing needed to attain such flatness would increase defects.

There are also overlay errors that can result from mask backside nonflatness, even if the frontside of a chucked EUV mask is flat (see Fig. 7.1). When the mask is chucked, there will be in-plane distortion that is proportional to the backside slope ϕ .^{2,3}

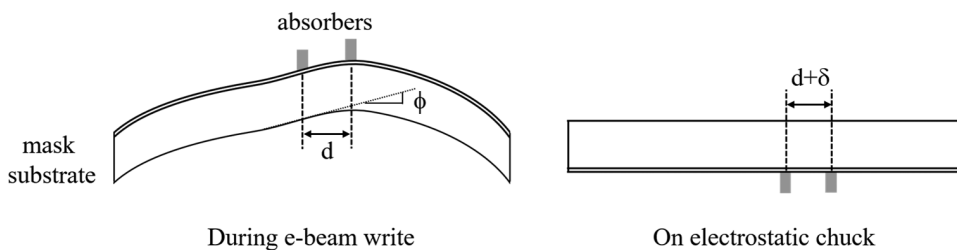


Figure 7.1 Mask shape during e-beam write and when on the electrostatic chuck.

Chapter 8

Metrology for EUV Lithography

Much of the metrology for EUV lithography is similar to that for optical lithography, with the caveat that EUV lithography is practiced at very small dimensions. Measurements of overlay and critical dimensions are needed, but since they have the same issues as they did for optical lithography, they will not be discussed in detail in this book. On the other hand, there are aspects of EUV technology without counterparts in optical lithography, mostly related to masks and pellicles, and these have special metrology requirements. Importantly, many metrology tools involve measurement at EUV wavelengths, which necessitates the use of specialized light sources and operation in vacuum.

Measurement tools are needed for the development and manufacturing of certain components, such as EUV mask blanks. These are of interest to practicing lithographers, even though such tools are not necessarily used in wafer fabs or masks shops, since it is often important for lithographers to understand the quality control capabilities of their suppliers. One of the earliest tools to be built in support of EUV technology development was a reflectometer at EUV wavelengths.¹ Such tools are used to measure multilayer reflectivity and uniformity on EUV mask blanks. Typically arranged to provide measurements of reflectivity at a 6° angle of incidence, the same as the chief ray angle of 0.33-NA exposure tools, these reflectometers can measure absolute reflectance over a range of wavelengths about 13.5 nm. These data can be used to determine the wavelength of peak reflectivity and the reflectance uniformity across mask blanks. Prior to the introduction of such standalone tools in 2002, measurements of reflectivity at EUV wavelengths were typically performed at electron storage rings. Standalone tools are much more convenient for manufacturers of EUV mask blanks. They can also be used to measure the impact of processing on multilayer reflectivity and changes in reflectivity over time. Standalone tools are currently available from EUV Technology, Inc.² and Research Instruments GmbH.³ Laboratories using synchrotron light sources continue to provide important calibration (as well as other) services.^{4,5,6}

Chapter 9

EUV Lithography Costs

Because the equipment and materials for semiconductor lithography are expensive, the overall cost of lithography has received considerable attention. There are a number of components that contribute to lithography costs (Table 9.1), and each of these will be discussed in more detail in this chapter as they pertain to EUV lithography. Because the impact of mask costs is highly dependent on the number of wafers produced per mask, wafer and mask costs are considered separately, although the situation for EUV lithography is somewhat more complex than for optical lithography, at least for situations in which pellicles are not in use.

Table 9.1 Categories for EUV lithography costs.

Wafer costs	Capital costs: tool prices, throughput, availability
	Maintenance costs
	Metrology, including mask qualification
	Operating costs: consumables, utilities
Mask costs	Blank costs
	Capital costs

9.1 Wafer Costs

9.1.1 Capital costs

EUV exposure tools cost in excess of \$120M.^{1,2} In order to realize a positive return on investments in such equipment, a high level of productivity per tool is necessary. The basic capital cost per wafer exposed is

$$\text{cost/wafer} = \frac{C_{ED}}{T_p U}, \quad (9.1)$$

where C_{ED} is the capital depreciation per hour, T_p is the raw throughput of the system (in wafers per hour), and U is the fractional equipment utilization.

Chapter 10

Extending EUV Lithography

Lithographic technology has been extended for several decades through a combination of reduced k_1 , increased numerical aperture, shorter wavelengths, and the use of multiple patterning. All of these are possible for EUV lithography and are the topics of this chapter. Because this chapter addresses future technologies that are still undergoing development, much of the text consists of descriptions of known problems that engineers need to solve, rather than explanations of solutions.

10.1 How Low k_1 Can Go

In the 1980s, it was conventional wisdom that k_1 needed to be 0.8 or larger in order to have a manufacturable process. Over time, improvements were made in resists, optics, and process control, and wavefront engineering was introduced, leading to useable values of k_1 for optical lithography below 0.3 and close to the physical limit of 0.25. For the full potential of EUV lithography to be realized, such low values of k_1 also need to be reached. EUV lithography has first been applied in high volume manufacturing to technologies with pitches ~ 40 nm, which represents a k_1 of 0.49, indicating that opportunities exist for the further reduction of k_1 , although there are challenges.

The fabrication of lenses with extraordinarily low aberrations was a significant reason that k_1 could be lowered for optical lithography. Aberrations on EUV exposure tools are below 0.2 nm rms.¹ Although impressive on an absolute scale, this level of aberrations is ~ 15 m λ , which is much higher than the levels achieved in optical exposure tools (< 4 m λ).² As a consequence, it will be difficult with EUV lithography to achieve the same low levels of k_1 reached with optical lithography. Nevertheless, with the level of aberrations existing in EUV lenses, substantial reductions below $k_1 = 0.49$ can still be expected. Further decreases in aberrations in EUV optics can also be expected, even if the same level of aberrations found in state-of-the-art immersion lenses is not achieved. Exactly how small a value of k_1 can be achieved will depend on how successfully the problems discussed in this chapter are solved.

Policy Brief

Tracing the Emergence of Extreme Ultraviolet Lithography

Lessons for Identifying,
Protecting, and Promoting the
Next Emerging Technology

Author

John VerWey

Executive Summary

This paper presents a case study on the most important technology to have emerged in the past decade: extreme ultraviolet (EUV) lithography. In 2019, when the first commercial electronics enabled by EUV were released, the technology was hailed as “the machine that saved Moore’s Law.”¹ All of today’s most advanced artificial intelligence (AI) chips, smartphones, autonomous driving systems, and high-performance computers contain semiconductors fabricated using EUV lithography. The Dutch company ASML has emerged as the sole supplier of EUV machines, winning a 30-year race that granted the company a monopoly on selling the tool essential for fabricating leading-edge semiconductors.²

However, while ASML gets well-deserved praise for developing and commercializing EUV, this paper’s focus is on the research community that supported EUV from the beginning: the academics in Japan, the United States, and Europe; the public-private partnerships; the conferences; and the industry collaboration that laid the groundwork for EUV in the 1980s and 1990s. Without this community, “the most technically advanced tool of any kind that’s ever been made” would not have been possible.³ This paper traces the academic, government, and industry actors involved in a multi-decade moon-shot project that ultimately saw EUV ascend from a speculative emerging technology to the mechanism that makes Nvidia’s leading-edge AI training chips and Apple’s latest smartphone possible.

Careful study of the research community that supported EUV development is particularly relevant for policymakers and the semiconductor industry today. EUV research began in the 1980s, when the U.S. semiconductor industry was trying to fend off ascendent Japanese firms amid significant government intervention on both sides. At the same time, the industry recognized that a new generation of lithographic light sources would be necessary to fabricate future advanced chips to maintain Moore’s law. Similar circumstances exist today, with policymakers in the United States, Europe, and Asia engaged in once-in-a-generation efforts to protect and promote their respective semiconductor industries, all while ascendent Chinese firms attempt to challenge industry leaders. Meanwhile, the entire semiconductor industry recognizes a slow-moving existential crisis: rapid advances in AI must be sustained by correspondingly rapid advances in computational power. However, the end of Moore’s law is in sight, and not even EUV can save it.⁴

The development of EUV reflects many of the emerging technology themes observed in previous CSET analyses.⁵ Research collaboration among academia, industry, and government has occurred for decades, making progress on associated technologies in

fields such as materials science, plasma physics, and chemistry. This progress has been documented in the form of journal article publications, patent filings, and conference proceedings. Over time, specific technologies would transition from government labs to the private sector, frequently through public-private partnerships and consortia formed to address and overcome technical hurdles. Eventually a private firm would assess a market opportunity, make investments to increase the technology's maturity, collaborate with customers, and commercialize the technology. Government support has always been essential at various points along the way.

Bibliometric study of the EUV research community during the period when EUV was still “emerging” offers important lessons for policymakers interested in identifying promising technologies today. It shows how academic research translates into scientific advances, the role of government and corporate labs, how international research collaborations accelerate innovation, the power of public-private partnerships, and the need for large amounts of patient private-sector capital. This paper catalogs this multi-decade path of innovation, identifying inflection points, signals, and sub-innovations along the way. Based on these findings, this paper concludes by introducing a set of criteria policymakers could use when attempting to identify future emerging technologies.

That a Dutch company, ASML, commercialized a technology pioneered in America's national laboratory ecosystem and largely funded by Intel also has important lessons for policymakers interested in protecting and promoting the next emerging technology.⁶ International collaboration on emerging technologies is inevitable, and guardrails on this collaboration come with trade-offs. Investments are an important source of funding for innovation, yet strategic acquisitions can fundamentally alter competitive dynamics in heavily consolidated industries. Finally, emerging technology supply chains evolve and mature in the private sector, whose incentives are profit and reliability, not geostrategic competitiveness. EUV tools are monopolized, but the EUV research community and underlying supply chain is globalized.

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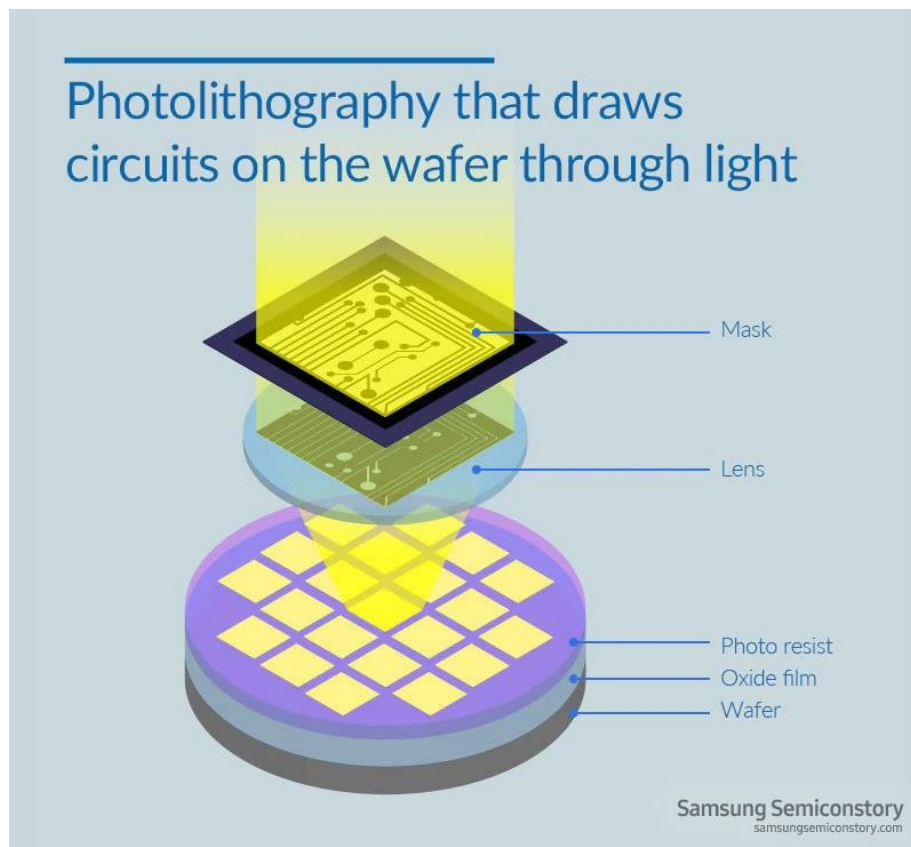
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Introduction

Background on Lithography and EUV

Extreme ultraviolet (EUV) lithography is the latest innovation in a long line of technical accomplishments that have supported the semiconductor industry's adherence to Moore's law from the 1960s to present.⁷ In semiconductor fabrication, lithography is the process of imprinting circuitry on silicon wafers using light-sensitive chemicals. The technology functions much like the process of developing a photograph: Light is filtered through an image (a "mask" in industry jargon) containing a circuit pattern. That pattern is projected onto the wafer, which is covered in light-sensitive chemicals ("photoresist"). The light interacts with the materials on the wafer surface, depositing the pattern as desired. This process is repeated dozens of times before the desired circuit pattern is fabricated on the wafer (Figure 1).

Figure 1. The Photolithography Process

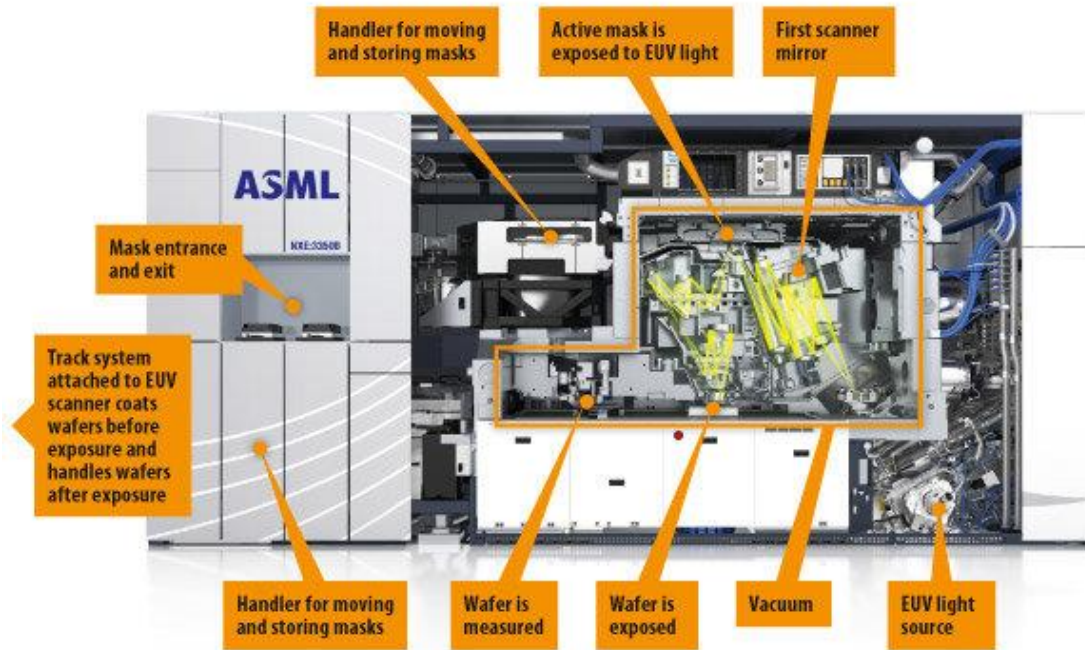


Source: Samsung, "Part 4, Drawing Structures in Nano-Scale," September 22, 2017, <https://semiconductor.samsung.com/support/tools-resources/fabrication-process/eight-essential-semiconductor-fabrication-processes-part-4-photolithography-laying-the-blueprint/>.

This lithographic process was first developed for use in the semiconductor industry at Fairchild Semiconductor and Texas Instruments in the late 1950s.⁸ Over time, as circuits shrank to the nanometer level, the industry was forced to adopt unique lithographic-specific light sources and associated technologies, because the visible spectrum of light exceeded the width of the desired circuit patterns. Specialty light sources were introduced, allowing for deep ultraviolet (DUV) lithography. However, the limits of DUV technology were well understood as early as the 1980s.⁹ Knowing the complexity of the task at hand, the semiconductor industry began to explore the scientific and technical merits of next-generation lithography (NGL) for fabricating advanced microelectronics decades before it would be necessary.

EUV lithography represents a departure from earlier photolithography approaches used by the semiconductor industry in several important respects, particularly the light source. EUV lithography functions by using high-powered lasers to incinerate 50,000 tin drops (each of which measures 30 micrometers, or 30-millionths of a meter) per second to generate extreme ultraviolet light.¹⁰ Because EUV light is absorbed by both air and glass, special mirrors, among the most flawless (as in, blemish-free) materials ever produced, then guide this light through masks and onto wafers in a vacuum chamber.¹¹ Ultimately, this process imparts the ultrafine circuitry necessary for today's most advanced semiconductors (Figure 2).

Figure 2. A Cross-Section Representation of an EUV Machine



Source: Maarten Steinbuch, Tom Oomen, and Hans Vermeulen, "Motion Control, Mechatronics Design, and Moore's Law," *IEEJ Journal of Industry Applications* 12, no. 2 (2022), www.jstage.jst.go.jp/article/ieejjia/11/2/11_21006010/_article.

Five thousand suppliers provide 100,000 parts, 3,000 cables, 40,000 bolts, and two kilometers of hosing to make an EUV tool. The tool weighs about 180,000 kilograms (200 tons), and ships in 40 containers spread over 20 trucks and three cargo planes.¹² ASML reportedly only makes 15 percent of the EUV tool in-house, partnering strategically with firms worldwide to source the highest quality components.¹³ This has resulted in a uniquely international product that is Dutch in name only. Though ASML does not talk in detail publicly about its suppliers and the EUV supply chain, Table 1 presents a list of recent known suppliers of specific EUV subcomponents.

Table 1. Summary of Notable EUV Subcomponents and Suppliers

Component	Subcomponent	Supplier(s)
Mask Blanks	Deposition	Veeco (U.S.), Applied Materials (U.S.)
	Inspection	KLA (U.S.), Lasertec (Japan)
	Blanks	AGC (Japan), Hoya Group (Japan)
Mask Patterning	Patterning	IMS Nanofabrication (Austria), NuFlare Technology (Japan), JEOL (Japan)
	Etching	Applied Materials (U.S.), Tokyo Electron (Japan)
	Cleaning	SUSS MicroTec (Germany), Applied Materials (U.S.), SCREEN (Japan)
	Inspection	KLA (U.S.), HMI (U.S., owned by ASML), NuFlare Technology (Japan), Lasertec (Japan)
	Defect Review	Lasertec (Japan), Zeiss (Germany)
	Repair	Zeiss (Germany), RAVE (U.S., owned by Bruker), Hitachi (Japan)
Mask Handling	Pellicles	ASML (Netherlands), Teledyne DALSA (Canada), S&S Tech (South Korea)
	Mask Pod	Entegris (U.S.), Gudeng Precision (Taiwan)
	Coaters and Developers	Tokyo Electron (Japan), Lam Research (U.S.)
Photoresist	Photoresist	JSR (Japan), Shin-Etsu Chemical (Japan), Tokyo Ohka Kogyo (Japan), Inpria (U.S., owned by JSR), DuPont (U.S.), Lam Research (U.S.), Gelest (U.S.)
Optics	Mirrors	Zeiss (Germany), Berliner Glas (Germany, owned by ASML)
Light Source	Laser	Cymer (U.S., owned by ASML), TRUMPF (Germany)
	Vessel	VDL Enabling Technologies (Netherlands)
Wafer Handling	Wafer Handling	VDL Enabling Technologies (Netherlands), Berliner Glas (Germany, owned by ASML)
	Metrology	HENSOLDT (Germany), ASML (Netherlands)

Source: Author's compilation.¹⁴

EUV tools are so essential to the semiconductor industry that ASML has become a technology industry leader: companies that can afford an EUV tool, which reportedly costs \$350 million as of 2023, are able to make the most advanced chips in the world.¹⁵ Companies that cannot afford EUV tools (or must wait while the back orders are cleared) settle for making second-best chips. This role has also put ASML in the middle of geopolitical tensions. Chinese firms, eager to make the world's most advanced electronics, are currently blocked from purchasing EUV tools on national security grounds by the Netherlands.¹⁶

The Development of EUV

The following sections describe the development of EUV from chronological and bibliometric perspectives. EUV development occurred in academic labs, corporate

research and development (R&D) facilities, private firms, and public-private partnerships over the course of 30 years. First, the chronological history of the EUV development program is presented, highlighting key contributions from academia, government, and industry, as well as the collective technical obstacles they worked to overcome.

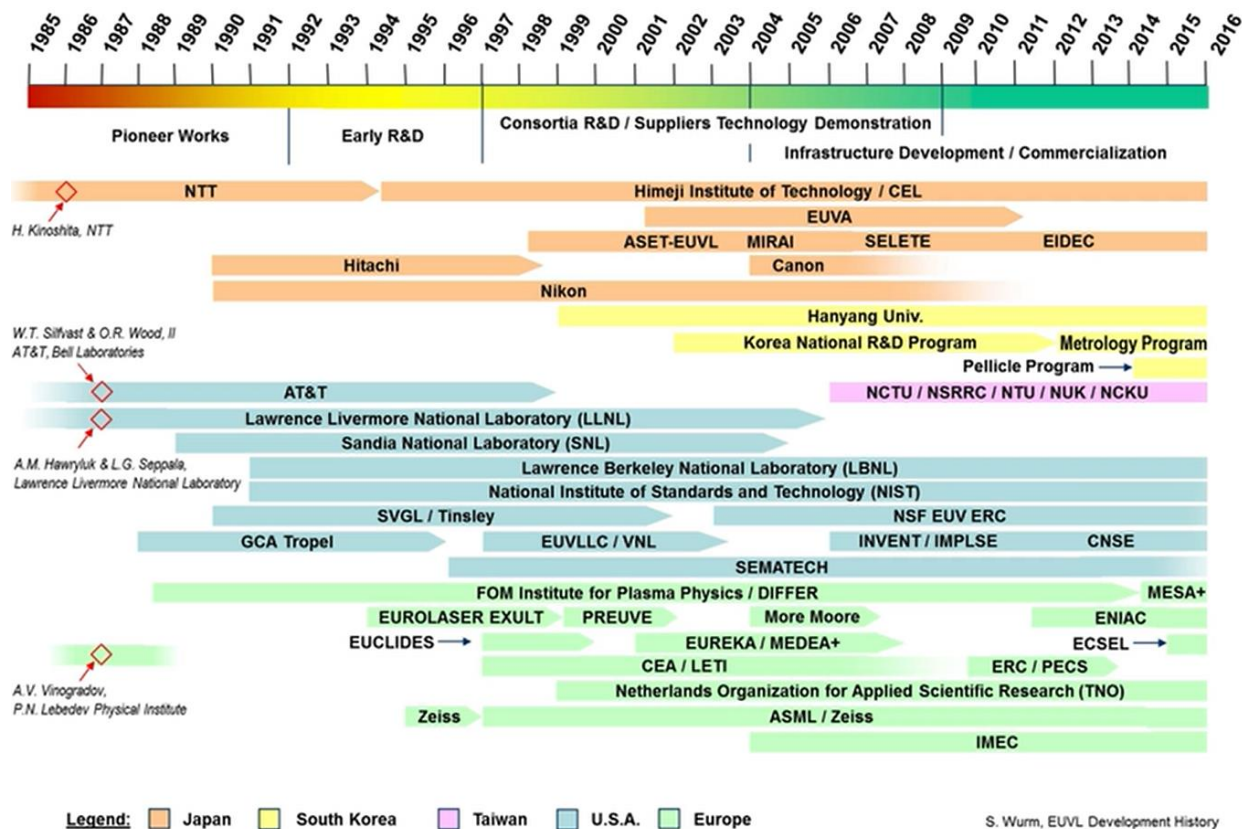
Second is a bibliometric analysis of the EUV research agenda over the course of 30 years. Much of EUV lithography's development occurred in the public record in the form of publications, conference proceedings, and patent filings. The analysis is derived from a book by the Society of Photo-Optical Instrumentation Engineers (SPIE), *EUV Lithography*, published in 2008 and revised in 2018.¹⁷ This manual presents a history of EUV's development from a technical perspective and was largely written by the key contributors active in the research community from the 1980s to the present. Importantly, it provides a comprehensive list of the consequential publications, actors, and narrative history.

This bibliometric analysis focuses on the EUV research community's accomplishments during the 30-year period when EUV was an emerging technology. The analysis demonstrates specific inflection points and accomplishments along the way that can be used as signals to identify other technologies that may be emerging today. It also reviews policymaker options for protecting and promoting emerging technologies at these inflection points.

EUV Origins

The first period of EUV research was focused on demonstrating the technical promise and feasibility of using soft X-ray projection lithography (SXPL) in semiconductor fabrication. The next period focused on developing the imaging systems and system integration, and identifying technical barriers and potential “showstoppers.” From the late 1990s through 2010, industry stepped in and focused the EUV research agenda on system development in preparation for high-volume manufacturing (Figure 3). From 2010 to the present, ASML has emerged as the sole supplier of EUV systems, successfully shipping its first allegedly “production-ready” EUV lithography tools in 2013. However, it was not until 2019 that the first electronic devices containing EUV-enabled semiconductors became available.

Figure 3. Timeline of EUV Development and Notable Actors



Source: Nelson M. Felix and David T. Attwood Jr., “EUV Lithography Perspective: From the Beginning to HVM (Conference Presentation),” SPIE Advanced Lithography, San Jose, California, 2020, video, 75:00, April 28, 2020, www.spiedigitallibrary.org/conference-proceedings-of-spie/11323/2572271/EUV-Lithography-Perspective--from-the-beginning-to-HVM-Conference/10.1117/12.2572271.full#_=_.

Fundamental Research and Technology Demonstration (1981–1992)

Soft X-ray reduction imaging, the technology that would later become EUV, was an idea ahead of its time. When, in 1981, researchers in the United States first started investigating whether soft X-rays could be used to impart images, their efforts focused on applications related to X-ray microscopes and X-ray telescopes.¹⁸ In fact, some of the very first tests evaluating soft X-rays for imaging were conducted at the Marshall Space Flight Center in Huntsville, Alabama.¹⁹ At this time, the leading NGL technology was X-ray proximity lithography (XPL). IBM in the United States and NTT in Japan were actively evaluating XPL for semiconductor device production, though both companies would later abandon the technology once manufacturing problems with XPL masks were deemed unresolvable.²⁰

Hiroo Kinoshita, an employee at NTT who emerged as a leading figure in the EUV lithography community, first began considering the applicability of this pioneering research for semiconductor device fabrication in 1984.²¹ Calling his approach X-ray reduction lithography, Kinoshita presented his initial findings at the Japan Society of Applied Physics in 1986 to little acclaim: the audience “seemed unwilling to believe that an image had actually been made by bending x rays.”²² Researchers in the United States experienced similarly negative feedback initially. Employees at AT&T Bell Laboratories approached the U.S. government in 1986, suggesting that soft X-ray lasers, paired with multilayer reflectors, could serve as a practical NGL tool. The proposal they prepared for the U.S. government received an “extremely negative” review, with the reviewers going so far as to say, “The printing of 0.1 [micron] features...would never be needed for [silicon] integrated circuits.”²³

By 1988, researchers at Lawrence Livermore National Laboratory (LLNL), part of the U.S. Department of Energy (DOE) national laboratory system, had picked up on Kinoshita’s research and proposed the first SXPL system.²⁴ These researchers, all of whom were affiliated with LLNL’s laser fusion program, took this pioneering research one step further, fabricating components and developing techniques to conduct diagnostics to verify SXPL’s promise.²⁵ Kinoshita’s research aligned with these efforts, and in 1989 he published a paper that proposed optimal SXPL exposure wavelengths, photomasks, and photoresist.²⁶ By the early 1990s, researchers at Sandia National Laboratories (SNL), another of the DOE’s national labs, had partnered with AT&T Bell Labs to demonstrate the first SXPL system with a laser plasma source. Momentum in the United States continued to grow as the U.S. Defense Advanced Research Projects Agency (DARPA) kicked off its advanced lithography program in 1991.²⁷ Soon thereafter, Kinoshita and several Japanese colleagues presented a refined, and

technically promising, two-mirror imaging system using SXPL in partnership with Tinsley Laboratories, a U.S. firm responsible for the optics used in the Hubble Space Telescope.²⁸ Finally, in 1992, Intel CEO Andy Grove approved a \$200 million company-internal R&D investment into EUV lithography.²⁹

Applied Research and Technology Development (1993–1997)

The second phase of R&D began with a name change: in 1993 the SXPL research community decided to coalesce on the term “extreme ultraviolet lithography.” This decision had several motivations. First, the term EUV distinguished this approach from the XPL research still being explored by IBM and NTT at the time.³⁰ Additionally, EUV sounded like an extension of DUV, the most common form of optical lithography then in use (Appendix 2 has more information on each of these lithographic approaches).³¹ The adoption of the term also served to remove the association with X-ray lithography (which didn’t work) and create an association with DUV lithography (which did work).³²

The U.S. National EUV Lithography Program emerged in 1994. Consisting of researchers from LLNL, SNL, Lawrence Berkeley National Laboratory (LBNL), and AT&T Bell Labs, this program was funded by DOE and steered by a technical advisory group with DARPA, DOE, and industry representatives. Teams at LLNL and SNL began developing imaging systems and the first tool leveraging EUV technology capable of precise overlay.³³

Development of mirror imaging systems (the “optics”) was a major focus of research during this period, with teams at SNL, Tropel Corporation, AT&T Bell Labs, and Silicon Valley Group (SVG) Lithography Systems all contributing to advances in the United States.³⁴ Related research in Japan continued to be led by NTT, with increasing participation from Hitachi and Nikon.³⁵

European research into EUV lithography also began in earnest during this period. Notably, an EUV research program called EXULT—which counted among its members ASM Lithography, the FOM Institute for Atomic and Molecular Physics, Sopra, Carl Zeiss AG, and Delft University of Technology—began a research agenda focused on “high-power debris-free [light] sources, precision optics fabrication and metrology, and defect-free masks.”³⁶ Europe’s first EUV workshop was organized and hosted by Zeiss in 1995.³⁷ The concentration of EUV research in the Netherlands at this time was notable. The Netherlands was seen by both the U.S. and Japanese semiconductor industries, which were at the time engaged in fierce competition, as neutral ground.³⁸ As a result, companies in Japan and the United States collaborated closely with

European firms. This head start would serve as a boost to an emerging Dutch company then known as ASM Lithography (later ASML).

Consortia and Industry-Led Technology Maturation (1997–2009)

By the mid-1990s, the International Technology Roadmap for Semiconductors forecast that a new lithography technology would need to be ready for high-volume manufacturing by 2005 if the industry was to keep pace with Moore's law.³⁹ However, despite years of exploratory work into XPL (led by IBM in the United States), projection electron and ion beams (led by AT&T Bell Labs in the United States), and shorter DUV wavelengths, no promising candidate technology had emerged.⁴⁰ At this inflection point, DOE's funding for the National EUV Lithography Program ended. This decision, the result of broader DOE budget reductions in the 1990s, occurred at a time when it was clear to industry that, if anything, the EUV lithography research agenda required ever greater funding.

The decision to cease funding the National EUV Lithography Program prompted Intel to step in and provide "bridge funding" to keep the U.S. EUV R&D community together until a more extensive program could be developed.⁴¹ In 1997, EUV LLC, led by Intel, was formed to push this research agenda forward in the United States. EUV LLC quickly established a contract with DOE through a novel national virtual laboratory (NVL) structure to ensure close alignment with the research teams at LLNL, LBNL, and SNL already engaged in EUV research.⁴²

Following the creation of EUV LLC in the United States, related consortia in Japan and Europe soon appeared. In 1998, the Japanese Association of Super-Advanced Electronics Technologies (ASET) established an EUV research agenda, and Europe's Extreme UV Concept Lithography Development System (EUCLIDES) was formed.⁴³ Unlike the U.S. government, the Japanese government and the European Commission provided direct financial support to these EUV research agendas.⁴⁴

Collaboration: Consortium Formation and the Global Research Agenda

With the creation of these public-private consortia in the United States, Europe, and Japan in 1997 and 1998, the semiconductor industry began to pursue EUV research in a much more concerted manner. Each of these consortia competed and collaborated in their efforts to commercialize EUV lithography systems. Recognizing the magnitude of the tasks at hand, the consortia established multiphase research agendas. While there was substantial overlap among the technical research agendas of the three consortia,

each of them expected that production-ready tools would not be available until the mid-2000s at the earliest.

Beginning in 1999, an annual international EUV workshop was held, rotating between the United States, Japan, and Europe. The function of this workshop, as well as other industry conferences and exchanges, served to advance overall progress on EUV research by sharing the results of their Phase I efforts (described below). Each consortium expected that Phase I would take three years and follow-on efforts aimed at developing a commercial tool would take a minimum of another three years, with EUV LLC aiming for a fab-ready system “on the floor in 2004.”

Japan, ASET, and EUVDA

Japan’s EUV research program at ASET conceived of a nine-year research agenda culminating with commercial tools ready for use in the mid-2000s. The program focused on three lines of effort in Phase I: the exposure system (including mirrors, the light source, and the mechanical system), multilayer masks (including mask patterning and defect detection), and photoresist development. ASET’s research agenda was coordinated by Hiroo Kinoshita (who by this time had departed NTT for the Himeji Institute of Technology), with participation from 10 semiconductor manufacturers and two equipment suppliers.⁴⁵

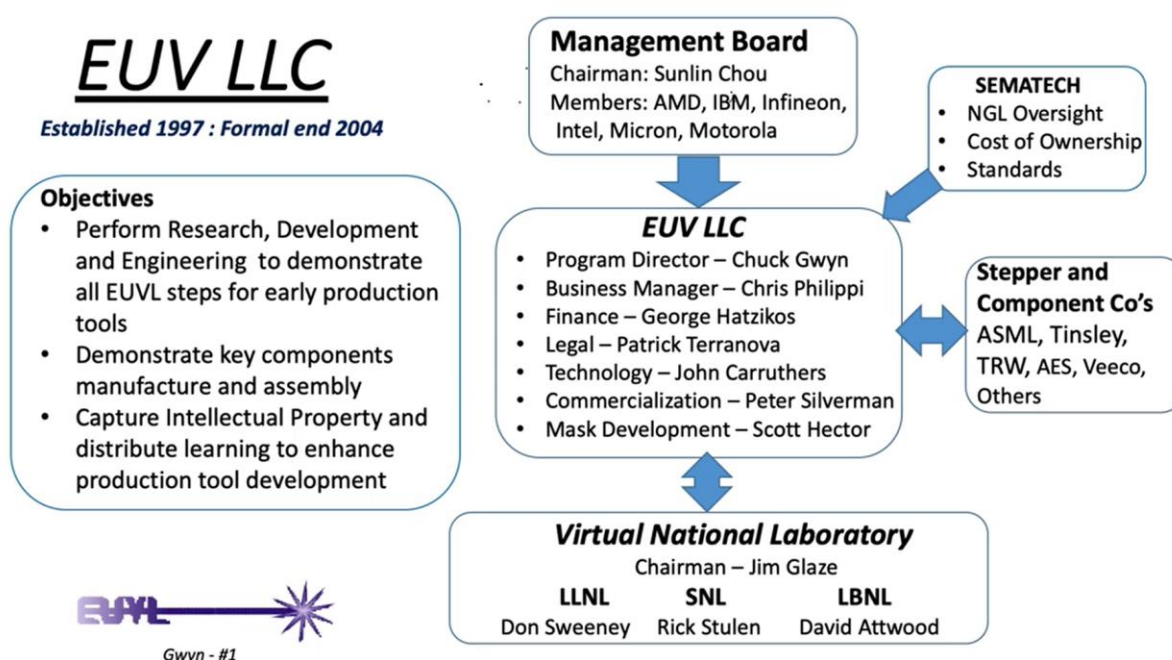
In some respects, Japan had a head start in NGL research: Japanese firms Canon and Nikon, who led in worldwide lithography sales in the mid-1990s, were pursuing company-internal NGL development efforts as well.⁴⁶ Canon had invested heavily into XPL, while Nikon had bet on electron-projection lithography (EPL). However, by late 2000, Canon had announced it would cease its XPL research in 2001.⁴⁷ The company argued that this line of research could be repurposed, giving it a head start in EUV work, while Nikon continued investigating EPL and EUV simultaneously.⁴⁸ Japanese firms and consortia continued to experience halting progress in EUV and other NGL technologies, which ultimately encouraged closer collaboration and technical exchanges between ASET and EUV LLC in 2001.⁴⁹

The United States and EUV LLC

The EUV LLC research agenda was the most well funded and ambitious of the three consortia. In Phase I, EUV LLC identified EUV optics, multilayer coating, metrology, masks, light source, photoresist, and development of an engineering test stand (a prototype EUV system) as “major tasks.” More important than funding was the level of participation from industry during this period: in addition to the aforementioned DOE

labs and Intel, ASML, Motorola, AMD, Micron, Infineon Technologies AG, and IBM ultimately joined EUV LLC (Figure 4).⁵⁰ No company was more committed to EUV research during this period than Intel. Preliminary estimates from Intel and the national lab partners in EUV LLC put an annual budget at around \$60 million, a number that would swell into the billions in the coming decade as Intel committed “what felt like infinite money for solving an impossible problem.”⁵¹

Figure 4. EUV LLC Structure



Source: Felix and Attwood, “EUV Lithography Perspective,” 37:30.

Intel and its EUV LLC partners ended up committing \$250 million to the DOE labs in 1997. Remarkably, Intel agreed to pay 100 percent of the national labs’ EUV research costs through a Cooperative Research and Development Agreement (CRADA) with the NVL that was hailed by the then secretary of energy as “the biggest investment the private sector has ever made in the Department of Energy.”⁵² Intel’s return on this investment was access to the 100-plus researchers in the DOE national labs engaged in EUV work, the rights to all EUV-affiliated intellectual property (IP) generated under the auspices of EUV LLC, and the (hopeful) continuation of Moore’s law.⁵³ DOE retained royalty-free right to use EUV-affiliated technologies generated by EUV LLC for defense applications.⁵⁴

Europe, ASML, and EUCLIDES

Led by ASML in partnership with Carl Zeiss AG and Oxford Instruments, Europe's EUCLIDES program focused its research efforts on mirror substrates, high-reflectivity multilayer coatings, and vacuum stages, while also evaluating plasma and synchrotron light sources.⁵⁵ EUCLIDES' progress in Phase I required access to industry know-how, applied R&D, and government user facilities and test beds. ASML chose to partner with TNO TPD, a Dutch institute for applied R&D; FOM Rijnhuizen, a Dutch research institute for plasma physics; PTB, the German metrology institute; and Fraunhofer IWS, a German institute for applied R&D focused on materials science and lasers.⁵⁶

The European research agenda into EUV was relatively small compared to the U.S. and Japanese efforts, yet it showed early signs of progress. Notably, Zeiss (the firm that today supplies all of the optics for ASML's EUV tools), in collaboration with Fraunhofer IWS, started to see early breakthroughs on the multilayer coating challenges in 2000.⁵⁷ Early on, ASML recognized it would need to rely on an extensive supply chain for many of the components and subcomponents as opposed to doing everything in-house.⁵⁸ Based on EUCLIDES' initial progress, in 2001 ASML "assigned a small group of people and modest resources to build a prototype EUV system."⁵⁹ This goal, which took five more years to realize, was aided by ASML's partnership with EUV LLC beginning in 1999.

Competition: Emerging EUV Leadership

During this period, in addition to the cooperative research agenda outlined in the previous section, there remained intense competition between countries, NGL technologies, and the companies championing them.

Competing Countries

The first question the secretary of energy fielded after announcing the formation of EUV LLC in 1997 was from a reporter who asked what steps the DOE was taking to ensure "American equipment manufacturers will benefit from this program and that it won't serve to perpetuate the dominance of Japanese equipment manufacturers in lithography technology."⁶⁰ At the time, the Japanese semiconductor industry was ascendent, and U.S. semiconductor firms were keen to maintain (and in some cases regain) their leading position. The U.S. government had anticipated this potentially controversial point and included an article in the agreement governing EUV LLC that established a "net U.S. benefit" clause, essentially requiring that participants commit to using EUV LLC-derived IP in support of U.S. manufacturing. Writing retrospectively,

the former head of EUV LLC stated, “The early failure to recognize that the U.S. government would prevent the EUV LLC from licensing lithography suppliers in Japan prevented the development of a robust competitive environment for [EUV].”⁶¹ Nevertheless, the industry recognized that the scale of the technical challenges necessitated closer international coordination. With Japanese collaboration politically unpalatable, EUV LLC looked to European partners.

In 1999, EUV LLC entered into an agreement with ASML that formalized the company’s participation in the consortium, closely aligning the work of EUCLIDES with U.S. efforts.⁶² Inclusion of this non-U.S. member firm was motivated by Intel, which had pushed for ASML’s participation as a means of accelerating research progress.⁶³ ASML’s inclusion in EUV LLC came over objections from some in the U.S. government who expressed concern that including non-U.S. partners in the LLC could result in the transfer of taxpayer-funded IP to a non-U.S. company. In exchange for membership in EUV LLC, ASML promised to build a factory in the United States comparable in scale to its Netherlands facility, agreed that 55 percent of the components in its future commercial EUV systems would be sourced from U.S. suppliers, and committed to sharing EUV technologies derived from the EUCLIDES program.⁶⁴ Soon thereafter, German firm Infineon Technologies AG also negotiated access to EUV LLC, offering to pay \$10 million, over objections from the U.S. Department of Commerce.⁶⁵

Competing Technologies

At least four NGL technologies were seriously considered by the semiconductor industry and championed by specific firms before EUV emerged as the clear winner (see Appendix 2 for more information). Each of these NGL approaches faced technical and economic obstacles. A complete lithography system requires (1) the exposure tool; (2) mask making, mask equipment, and mask materials; (3) the photoresist materials and processing equipment; and (4) metrology equipment for critical dimension and overlay.⁶⁶ Significant technical innovation was required for each of these subcomponents and functions. In almost all cases, though, economic considerations dictated the rise and fall of different NGL approaches. Ultimately, these tools needed to be able to conduct hundreds of wafer exposures per hour (referred to as “throughput”), with as little downtime as possible, for years at a time. NGL technologies that lacked throughput and reliability did not prove economically viable.

SEMATECH, a nonprofit consortium founded in 1987 to sustain U.S. semiconductor industry leadership, convened an NGL task force in 1997 that identified XPL, EPL, and ion projection lithography as the most likely candidates (each are described further in Appendix 2).⁶⁷ EUV came in fourth place. Nevertheless, ASML’s participation in EUV

LLC in 1999 accelerated enthusiasm for EUV lithography at the expense of the other lithographic approaches then under consideration. Less than two years later, in early 2001, ASML dissolved eLith LLC, a joint venture with U.S. firm Applied Materials that had been focused on developing a next-generation EPL tool derived from research at AT&T Bell Labs.⁶⁸ In making the decision to abandon EPL, ASML cited EUV's promise as a "more extensible" technology—offering manufacturing capabilities for multiple generations of processor technology—as well as EPL's relatively lower throughput.⁶⁹

In 2002, ASET reported an alpha prototype EUV tool capable of fabricating chips at the 35nm node.⁷⁰ That same year, Japanese firms, with the support of Japan's Ministry of Economy, Trade, and Industry, established the Extreme Ultraviolet Lithography System Development Association (EUVA).⁷¹ Membership in EUVA consisted of all the leading Japanese semiconductor firms: Canon and Nikon; light-source manufacturers Komatsu, Ushio, and Gigaphoton; and device manufacturers Fujitsu, NEC, Renesas Technology, and Toshiba.⁷² ASET focused on developing mask and resist technology, while EUVA and its leading members Nikon and Canon focused on optics and exposure tool ("scanner") development.⁷³ Despite making progress in these areas, by 2003 Japan's efforts in EUV were already being characterized as "play[ing] catch up."⁷⁴

Part of the reason Japanese firms were perceived to be falling behind in EUV was the success of U.S. efforts under EUV LLC. The consortium, which ceased operations in 2004, resolved many of the major tasks initially identified in 1997 and delivered the first prototype EUV lithography tool, referred to as the engineering test stand. Its six-year research effort is generally regarded as a success. Yet, substantial technical hurdles remained by the time of EUV LLC's dissolution, and its expected delivery of an alpha tool "on the floor in 2004" was now delayed several years further. At this point, SEMATECH stepped into the void left by EUV LLC. SEMATECH focused on EUV commercialization efforts, building infrastructure to support production of EUV mask blanks and photoresist, providing light-source data, and characterizing optics lifetime and contamination.⁷⁵

This delay did not cause an end to Moore's law as the International Technology Roadmap for Semiconductors had earlier predicted, primarily because of successful technical innovations that extended DUV lithography to fabricate all the way down to the 45nm node and below.⁷⁶ Using DUV lithography for more advanced device fabrication is slower and more resource intensive (more masks and photoresist are required, increasing costs and the possibility of defects) but technically viable. ASML now estimates that, while it takes only nine EUV lithography steps to fabricate a 7nm chip, DUV lithography tools can accomplish the same task in 34 steps.⁷⁷

As early as 1999, the industry recognized that DUV extension could be viable, and by the early 2000s DUV lithography's successful extension actually reduced the urgency of the EUV research agenda.⁷⁸ By the mid-2000s, only hard problems remained: it had become apparent to the industry that EUV technology would not be ready until the end of the decade, if ever. Key technical issues were not being resolved in a timely manner, delaying the project writ large. These issues included "[light] source output power and lifetime; availability of defect-free masks; reticle protection during storage, handling and use; projection and illuminator optics lifetime; resist resolution, sensitivity and line width reduction; and optical quality for the 32-nm node."⁷⁹ In 2005, these ongoing technical hurdles led several researchers to declare that EUV lithography was "headed for failure," with one keynote speaker predicting, "No one will ever print a wafer for profit with EUV."⁸⁰ This same year, DARPA ended funding for its advanced lithography program, estimating it had spent \$1 billion over the program's 14-year existence.⁸¹

Competing Companies

In 2000, ASML, then the second largest lithography company in the world, made an offer to acquire SVG Lithography, the fourth largest. The acquisition—which was approved in 2001 after an extensive national security review process in the United States that forced ASML to divest SVG's Tinsley Labs business—made ASML the largest lithography vendor in the world, putting it ahead of Canon and Nikon.⁸² Importantly, the purchase of SVG also gave ASML the inside track to win future business from Intel, which had historically relied on SVG for lithography tools.⁸³ With SVG's acquisition, however, the last American lithography-tool supplier ceased to exist.⁸⁴ Not long after, Northrop Grumman made an offer for TRW, a firm supporting EUV LLC's work on light sources, and, subsequent to the closing of the transaction, exited light-source development. Northrop Grumman ended up donating all of its EUV-affiliated IP to the University of Central Florida in 2004.⁸⁵

These acquisitions came at a fraught time for the semiconductor industry. Global semiconductor sales fell 32 percent following the dot-com crash in 2001 and recovered slowly from 2002 to 2005, hurting firm profitability and thus R&D budgets.⁸⁶ One bright spot was Intel, whose personal computer business proved resilient and exceedingly profitable. Intel used these profits creatively during the downturn, funding ongoing research into EUV at a time others could not afford to. In 2003, Intel purchased \$90 million worth of Nikon bonds, "partly to ensure Nikon climbed fully on board the EUV bandwagon."⁸⁷ It invested another \$20 million in a San Diego-based developer of EUV light sources called Cymer to sustain its operations.⁸⁸

ASML and Nikon competed head-to-head for lithography market share in the mid-2000s, simultaneously pursuing different NGL technologies.⁸⁹ Adding to this competition, Nikon sued ASML for patent infringement in 2001 and reached a settlement three years later, with ASML and Zeiss paying \$87 million and \$58 million to Nikon respectively.⁹⁰ In spite of this win for Nikon, that same year it became increasingly apparent that its big bet on EPL was never going to reach the throughput necessary to be competitive with EUV. In 2005, it ceased efforts to commercialize EPL.⁹¹

ASML, Nikon, and Canon continued to guarantee EUV alpha tools to customers by the mid-2000s, with production-ready tools by 2009.⁹² In 2006, ASML shipped its first EUV prototype tools to Belgium's Interuniversity Microelectronics Centre (IMEC) and the College of Nanotechnology, Science, and Engineering (CNSE) in New York State.⁹³ In 2007, Nikon reported that its first two EUV tools would be shipped to Intel and Selete (a Japanese joint venture).⁹⁴ In 2008, CNSE successfully used ASML's demo tool to produce the world's first EUV test chips.⁹⁵ Canon continued to struggle with EUV and by 2009 had decided to exit the race, leaving only ASML and Nikon as viable suppliers to the industry.⁹⁶ That same year, it was rumored that Nikon's EUV program was experiencing further delays.⁹⁷

Commercialization and High-Volume Manufacturing (2010–Present)

It was time to make money. ASML and Nikon had been taking orders for their long-delayed EUV tools for years, citing per-tool prices between \$15 million and \$20 million in 2001.⁹⁸ But in 2010, ASML's first preproduction EUV system was finally shipped to a Samsung research facility in South Korea, and the cost had risen to \$120 million per tool.⁹⁹ The costs of developing EUV had escalated to the point that Nikon elected to exit EUV lithography system development altogether in 2011, leaving ASML as the sole firm pursuing EUV.¹⁰⁰ Yet getting to this point had cost ASML tremendously, and it found itself strapped for cash just as the first EUV tools were being evaluated by its customers. In response, ASML announced a 2012 "customer co-investment program," in which its most important customers—Samsung, Intel, and TSMC—would fund further EUV research and rollout in exchange for equity in ASML.¹⁰¹ Intel committed \$4 billion, while Samsung and TSMC each committed \$1 billion.¹⁰²

With this assurance, ASML doubled down on its efforts to commercialize EUV tools. The last major technical hurdle to be addressed had to do with the light source. In late 2012, ASML made an offer to acquire Cymer, the San Diego-based DUV and EUV

light-source developer, for \$2.5 billion to vertically integrate its supply chain further and accelerate progress.¹⁰³

From this point forward, the story of EUV's emergence becomes much more well-known. The light-source challenges were resolved by ASML and Cymer, with critical help from the German laser manufacturer TRUMPF. In 2013, ASML shipped its first EUV production system, officially signaling the “emergence” of the technology, though it would not be ready for high-volume manufacturing in a leading-edge fab until Samsung and TSMC started shipping the first EUV-enabled products in 2019.¹⁰⁴ ASML's ascendance in the lithography world coincided with the decline of Japanese lithography firms, who today maintain about 10 percent of the worldwide lithography market, with ASML holding roughly 90 percent market share (and, of course, for EUV, 100 percent market share).¹⁰⁵

The EUV Research Community: Characterizing Signals of Emergence

Throughout the timeline of events outlined in the previous section, the research community engaged in EUV commercialization was actively publishing, patenting, investing, and partnering on affiliated topics. This corpus of work provides a rich case study for researchers interested in identifying emerging technologies. Importantly, much of this information has already been collected and organized thanks to SPIE's definitive *EUV Lithography* volume. Using this book, it is possible to define the members of the research community and the discrete tasks on which they worked and to conduct analysis of this output over time to look for signals of emergence.

The analysis provided in this section is derived from SPIE's *EUV Lithography*. SPIE conferences were among the most important for the EUV research community in the 1990s and mid-2000s and provided a forum for researchers to share findings internationally. This volume, first published in 2008 and revised and republished in 2018, provides a comprehensive history of EUV development and an overview of each major subsystem, along with the technical hurdles overcome. The narrative of events, actors involved, and publications cited in the section that follow are extracted from this volume.

Research Community Composition and Participation

Writing his history of EUV development from the 1980s through 1996, Hiroo Kinoshita lists about 300 researchers by name as having made “important contributions” to EUV lithography during this period.¹⁰⁶ This group, though not an exhaustive list, likely represents the most comprehensive snapshot of the EUV research community from the 1980s through mid-1990s. The United States (about 225 researchers), Japan (about 45 researchers), and Europe (about 30 researchers) were home to this research community.¹⁰⁷ Roughly 10 percent of researchers were affiliated with a university, 30 percent were affiliated with a government-supported research institution (i.e., a national laboratory), and the remaining 60 percent were employed in the semiconductor industry.¹⁰⁸

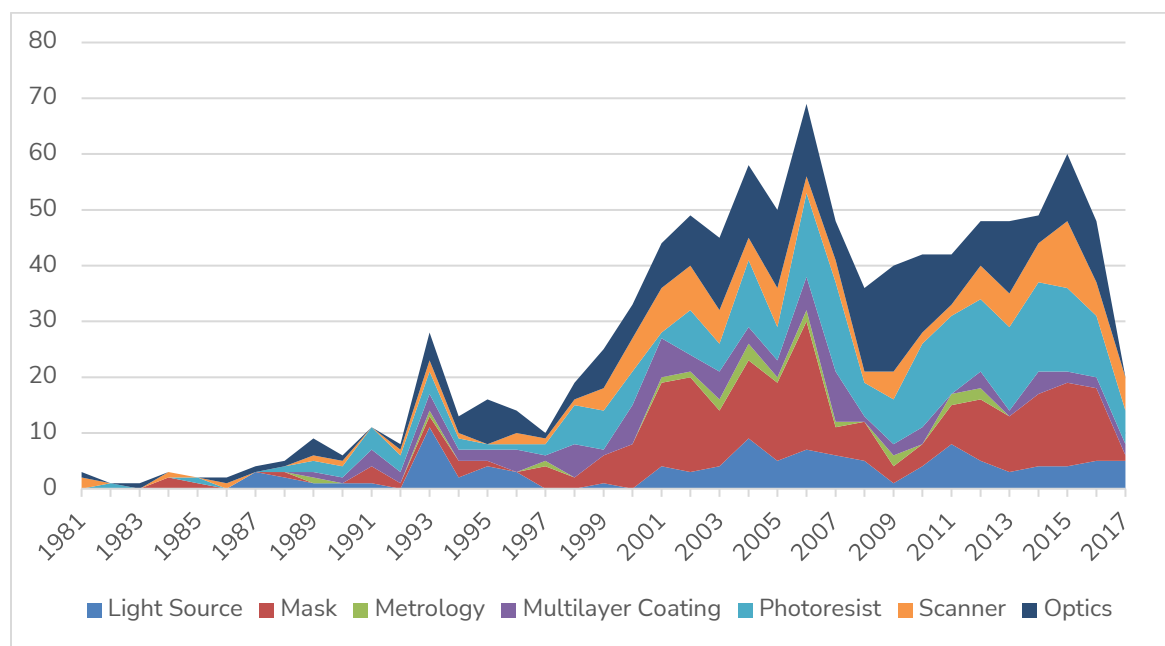
With the formation of the consortia in the late 1990s, the composition of this research community shifted decisively from academia to industry. The overall number of participants in Japan's ASET and EUVA research programs and in Europe's EUCLIDES program is more difficult to ascertain. However, over 2,800 individuals are listed as a coauthor on one or more publications in the SPIE's history of EUV, indicating the research community grew dramatically from the mid-1990s onward.¹⁰⁹ This number is likely a conservative representation of worldwide participation in EUV research, as

many key contributors were employed by the industry at the time and not publishing on their work.

Research Community Publications and Technical Areas of Focus

This research community publicly reported on progress and obstacles in the form of academic publications from the 1980s through the 2010s. Of the 1,035 distinct EUV-affiliated publications cited in the SPIE's volume, the vast majority of them were published from 1997 onward (Figure 5).¹¹⁰ Interestingly, there are two observable spikes in EUV-affiliated publications around 1993 (when the research agenda for EUV became more formalized) and again around 2006 (when ASML shipped its first alpha tools to IMEC and CNSE). Each of these periods was subsequently followed by a steep decline in publications, an EUV winter of sorts, when interest in (and perhaps enthusiasm for) the technology waned in the face of technical and economic obstacles.¹¹¹

Figure 5. Number of EUV-Affiliated Publications Annually, 1981–2017



Source: Author's compilation, derived from Hiroo Kinoshita and Obert Wood, "EUV Lithography: An Historical Perspective," in *EUV Lithography*, ed. Vivek Bakshi (Bellingham, WA; SPIE, 2008), www.spiedigitallibrary.org/ebooks/PM/EUV-Lithography/1/EUV-Lithography-An-Historical-Perspective/10.1117/3.769214.ch1.

Among the technical areas that saw the most sustained research and publication output over time, photomasks, photoresist, and optics stand out. Each of these areas presented significant scientific and technical obstacles to the overall research agenda. Interestingly, the EUV light source, which was the last and most difficult obstacle to overcome, has a comparatively small publication footprint.¹¹² This may be because the companies developing EUV light sources chose to keep that information as a trade secret during the period of development. This may also explain why EUV-affiliated metrology publications, a similarly remarkable technical accomplishment, are underrepresented.

The 1,035 publications cited in the SPIE volume represent a lower bound of the total EUV-affiliated publications from the 1980s to present. A simple query of “EUV lithography” in Scopus and Web of Science indicates a similar spike in interest from the 1997 period onward. However, Web of Science indicates an average of 248 EUV-affiliated publications per year (and a median of 267) from 1997 through 2017. This suggests that thousands of additional papers dealing with topics related to EUV lithography exist (see Appendix 3 for more information).

Identifying, Protecting, and Promoting the Next Emerging Technology

The story of EUV shows that early government interventions matter most with respect to promoting emerging technologies, while government interventions to protect such technologies either come too late or lack relevance due to a limited tool kit. Interestingly, the moments when governments can make the most difference in promoting an emerging technology are the same moments when governments are most powerless to protect it. Conversely, by the time a technology is mature or “ready” enough to merit protection, there is very little policymakers can do to continue to promote its development, as it will most likely have “emerged” in the hands of industry.

Government efforts to identify emerging technologies are necessary but insufficient to protect and promote them. Though it may have been possible for policymakers to spot the emergence of EUV using some of these signals, the simple reality is that emerging technologies start out as arcane ideas championed by a small cohort of intensely passionate researchers. In general, policymakers lack the technical literacy, resources, and patience necessary to characterize and track a technology’s emergence. Nonetheless, policymakers can use the template introduced in the next section to assist in identifying emerging technologies based on bibliometric and economic signals: publications and investments.

Identifying the Next Emerging Technology

Previous CSET research has described the challenges and opportunities policymakers face when attempting to identify emerging technologies.¹¹³ By definition, emerging technologies are new and rapidly changing, making them hard to track. Because these technologies are inherently immature, the underlying chain of people, processes, products, services, information, and resources that supports the technology is difficult to define. Additionally, technology development is diffuse and occurs in universities, firms, and government labs, often simultaneously and across borders.

“Emerging technology” as a concept also lacks an agreed-upon definition. For the purposes of this paper, an emerging technology is “a relatively fast growing and radically novel technology characterized by a certain degree of coherence persisting over time and with the potential to exert a considerable impact on socio-economic domain(s).”¹¹⁴ Using publication data and investment trends described earlier in the paper, Table 2 presents a template researchers and policymakers could use to identify other emerging technologies by quantitatively assessing coherence and growth and

qualitatively assessing impact and novelty (more context on each of these attributes is in Appendix 1).

Table 2. Signals of Emerging Technology Coherence, Growth, Impact, and Novelty

	Coherence	Growth	Impact	Novelty
Publications	• >100 contributors (representing 25+ academic, industry, and government institutions) • A total of 50+ publications annually for a five-year period	• Number of contributors doubles in a five-year period, exceeding 500 individuals • Number of publications increases 100% year over year for a five-year period, exceeding 100 publications annually	Leading researchers are listed as inventors on patent filings	A new annual conference series or journal emerges and remains in operation for five-plus years
Investment (R&D, capital expenditure, mergers and acquisitions, venture capital)	>\$100 million cumulatively invested in affiliated technologies	Value of investments doubles every other year (or year over year) for a five-year period, cumulatively exceeding \$250 million	• Investments drive consolidation or vertical integration • Incumbent firms make substantial (\$100 million+) investments in the technology	Government and industry coinvest to form public-private partnerships that explore technology maturation and commercialization

Source: Author's creation.

The development of EUV met these conditions at different points in time:

- **Novelty and coherence:** The EUV community exhibited nascent signals of coherence early on, during the 1989 to 1993 period. While only about 15 EUV-affiliated publications appeared annually on average during this time, Intel committed \$200 million to EUV research in 1992, the research community harmonized on the term “EUV” in 1993 at a conference convened specifically for the topic, and the research community actually exceeded 300 contributors by the mid-1990s. Governments also formed the first EUV-focused public-private partnerships to explore the technology in earnest during this time, notably the U.S. National EUV Lithography Program and Europe’s EXULT program.
- **Growth and impact:** The EUV community’s rate of publication and investment demonstrated significant growth from 1997 to 2003. The number of named contributors on papers, a proxy for the size of the research community, doubled during this period. Likewise, the overall number of EUV-affiliated publications increased by 100 percent year over year, breaking 50 publications for the first time in 1998 according to Web of Science data. EUV-affiliated investments by major actors—which include mergers and acquisitions as well as R&D—also grew dramatically. ASML purchased SVG Lithography during this period.

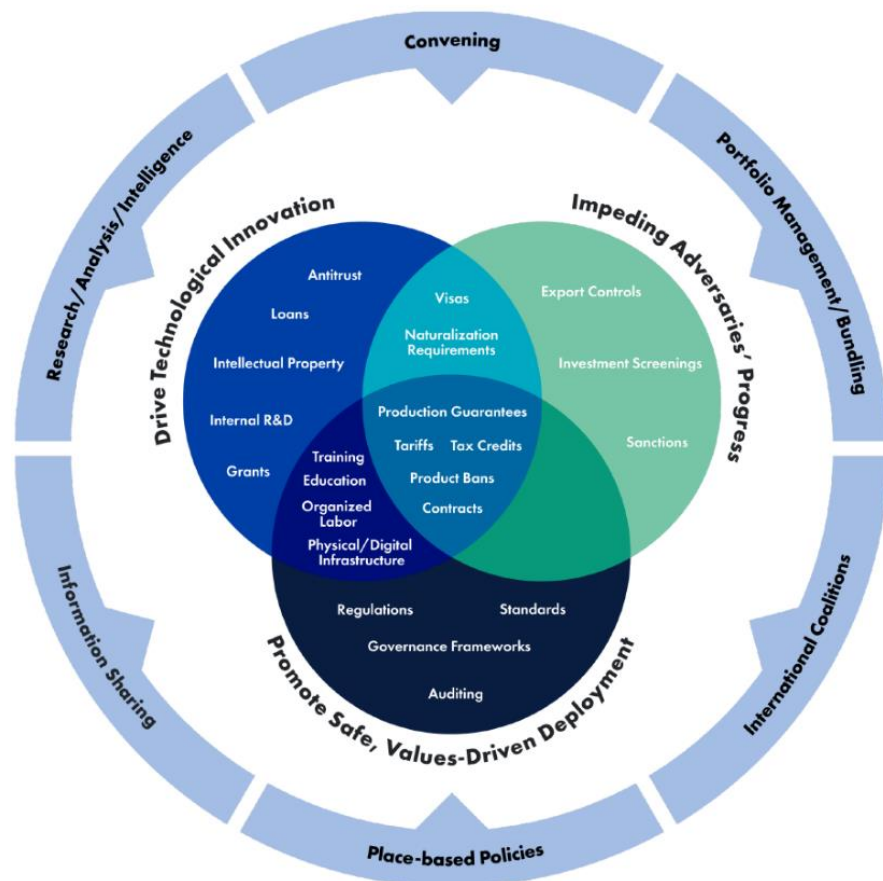
Protecting and Promoting the Next Emerging Technology

How did the most important emerging technology of the past 10 years, a technology that was largely pioneered in U.S. government labs and funded by Intel, end up monopolized by a Dutch company and sold to Intel’s largest competitors first? The short answer to this question is that ASML had first-mover advantages, pursued strategic acquisitions, was able to weather cyclical downturns, enjoyed remarkable direct financial support from key customers, made a high-risk bet on an unproven technology (unlike its more conservative Japanese competitors), benefited from bad bets and competition among its rivals, and remained focused on its customers. Additionally, Intel stumbled.¹¹⁵ Policy interventions had little to no impact on the timeline or maturation of EUV from roughly 2006 onward.

Answering this question more broadly has important implications for policymakers. Previous CSET research has detailed the trade-offs policymakers face when pursuing strategic technology and national security goals, especially with respect to driving technological innovation and impeding adversaries’ progress (Figure 6).¹¹⁶ The

development of EUV has several lessons for policymakers interested in promoting and protecting emerging technologies.

Figure 6. Policy Instruments and Enabling Levers



Source: Jack Corrigan, Melissa Flagg, and Dewey Murdick, “The Policy Playbook: Building a Systems-Oriented Approach to Technology and National Security Policy” (CSET, June 2023), <https://cset.georgetown.edu/publication/the-policy-playbook/>.

Promoting Emerging Technology

The story of EUV’s emergence contains themes shared with other research agendas. Technologies often take decades to “emerge.” These technologies grow, change, and mature in unexpected ways at a pace that either exceeds policymakers’ abilities to productively intervene or is so slow that policymakers lose interest. International collaboration is inevitable (and desirable) when advancing complex, multifaceted research agendas. Government support—including direct financial support for public-private partnerships, indirect financial support, and funding basic science user facilities—is essential. Equally essential is partnership with the private sector, whose

support (in the form of large amounts of patient capital, technical know-how, and supplier-customer feedback loops) is necessary for commercializing an emerging technology. Collaborative development (and licensing) of IP can accelerate technology timelines, as can competition among private-sector actors.

The history of EUV suggests government intervention to promote emerging technologies is most valuable at early stages. This intervention can take several useful forms: retaining subject matter experts at national research facilities, endowing scientific user facilities with the resources necessary to pursue speculative high-risk high-reward research agendas, ensuring these facilities are open to international collaboration (either through direct access or the licensing of IP), direct financial support, creating public-private partnerships, facilitating information sharing, and convening otherwise-disparate research communities around specific topics for conferences or road mapping exercises.

With the benefit of hindsight, the single best policy the U.S. government pursued to promote EUV development was allowing national laboratory researchers engaged in EUV research to work with EUV LLC under the auspices of the virtual national lab from 1997 to 2003. However, EUV LLC only came into existence because Intel could afford to foot the bill at the time. Sustained government investment in basic and applied science user facilities also promoted EUV research. Importantly, these facilities convene academia and industry, remain open to worldwide collaboration, and offer no-cost joint development projects to startups with strict IP protections and rights afforded to all participants. Finally, progress on large-scale scientific research agendas (such as those scoped by the National EUV Lithography Program, ASET, EUCLIDES, and EUV LLC) work best when given large amounts of patient capital: \$200 million to \$300 million per year for a five-to-ten-year period.¹¹⁷

Protecting Emerging Technology

The primary mechanisms any government may use to protect a technology (research security, investment screening, export controls, and supply-chain security) are challenged by the nature of emerging technologies. Though governments can limit a technology's spread through guardrails that inhibit international collaboration or access to IP, stipulate who can and cannot invest in emerging technology firms, stop or interdict the international transfer of a technology, and assure strategically advantageous technology supply chains, emerging technologies are not so easily protected. Government technology protection regimes generally focus on ensuring economic or national security, and emerging technologies rarely present clear implications for these in their initial stages.

In some respects, the U.S. government made two key choices to “protect” EUV at a relatively early stage in its development via EUV LLC. First, the U.S. government effectively blocked Japanese lithography firms from direct participation in EUV LLC. Second, it permitted ASML’s participation in EUV LLC, pursuant to provisions stipulating that ASML guarantee a certain amount of U.S.-based manufacturing and U.S.-origin content in its EUV tools. This latter requirement in particular guaranteed that the tools would contain enough U.S.-origin content to meet U.S. export-control thresholds. However, both of these decisions were not made by the U.S. government with explicit technology protection in mind but were rather a function of political pressure to ensure that taxpayer dollars would not be used to develop a technology that would ultimately be commercialized overseas.

The history of EUV shows that international collaboration and investment are essential and the ability to transfer technologies across borders is necessary for commercial sales as well as to source the highest quality components. Government efforts to protect emerging technologies run the risk of slowing down research and limiting firm profitability (and thus companies’ ability to reinvest those profits in future R&D) at a time when a technology does not yet present clear economic or national security implications. Because nearly every emerging technology today is “dual use” (that is, it has commercial and potential military applications), efforts to protect these technologies run counter to the desires of the companies actively developing them in response to a perceived commercial opportunity.

This was particularly true in the case of EUV. Policymaker efforts to protect U.S.-origin innovations around EUV would have hampered overall development and disadvantaged U.S. firms such as Intel who were depending on the innovation. In fact, the primary reason ASML ended up in a monopoly position with respect to EUV technology is because of Intel. Intel assessed that SVG Lithography could not be relied on to supply an EUV system and therefore lobbied for its acquisition by ASML. Intel also argued for ASML’s inclusion in EUV LLC. And Intel bet on ASML over its Japanese lithography rivals. U.S. efforts to protect EUV as an emerging technology would have run counter to Intel’s stated goal: whoever could get EUV to market fastest was its preferred supplier, and only ASML fit the bill. Unfortunately, by the time EUV was ready for high-volume manufacturing, Intel’s fabrication stumbles in transitioning from 14nm to 10nm and then 7nm semiconductors meant that EUV adoption actually added to its challenges rather than solved them. Only in 2023, five years after EUV’s introduction in high-volume semiconductor manufacturing, did Intel starting to make use of the technology it first bet on 30 years ago.¹¹⁸

Conclusion

Like many other emerging technologies, EUV development required persistent collaboration and innovation between academia, industry, and government. This process is inherently global, resulting in a constant tension between government efforts to cooperatively promote and competitively protect a high-value technology. Promoting a technology requires early and sustained intervention and necessitates creative partnerships with industry. Protecting an emerging technology from specific competitor countries risks undermining innovation and frequently runs counter to the goals of the companies tasked with commercialization.

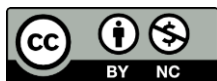
As a technology matures, government and industry priorities inevitably diverge: industry-led technology commercialization results in global supply chains and customer relationships that prioritize profit and reliability. Government efforts to control emergent technologies run risks ranging from ineffectual interventions to slowing innovation. The case of EUV highlights that there are no solutions, only trade-offs, for policymakers who want to identify, protect, and promote the next emerging technology.

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Acknowledgments

For helpful feedback and suggestions, the author would like to thank Igor Mikolic-Torreira, Melissa Flagg, Autumn Toney, Taylore Roth, Kevin Boyack, Richard Klavans, Caleb Smith, Chris Miller, Chris Mack, James Dunham, and Rebecca Gelles. He would also like to thank Dale Brauner, Jahnavi Mukul, Shelton Fitch, Owen Daniels, and Liz Dana for editorial support.



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Document Identifier: doi: 10.51593/20240003

Appendix 1. Emerging Technology Attributes¹¹⁹

Characteristic	Concept	Example
Radical novelty	Innovation that is clearly differentiated from past technologies	Wireless communication technologies
Fast growth	An increase in the number of actors, funding, or outputs	Quantum computing
Coherence	Convergence of previously separated research streams	CRISPR-based gene editing
Prominent impact	Potential to create an entirely new industry or fundamentally change an existing one	Google's PageRank algorithm
Uncertainty or ambiguity	A variety of potential applications, but high-confidence future predictions are difficult	Biofuels

Appendix 2. Next-Generation Lithography and Light Sources

Today's most advanced microelectronics have nearly atomic-level features. The ability to fabricate these microelectronics is a credit to EUV, and especially to the light source used in the system. The smaller the wavelength of light used to fabricate a chip, the smaller the features of that chip. As a result, the light source emerged as the single most challenging and consequential aspect of the entire NGL research agenda. This appendix summarizes the types of NGL technologies that were considered by industry from the nineties to the aughts, as well as the types of lithography light sources used by industry from the 1970s to present.

As discussed earlier in this paper, developing EUV required overcoming technical and economic obstacles. With respect to the light source used in EUV, this meant that not only did a new way of generating EUV light need to be invented, but the technology also needed to produce chips at volumes that made it profitable. This more or less doomed each of the other competing NGL technologies and light sources under consideration, as they were too expensive, too slow, or too complicated to operate reliably (Table 3).

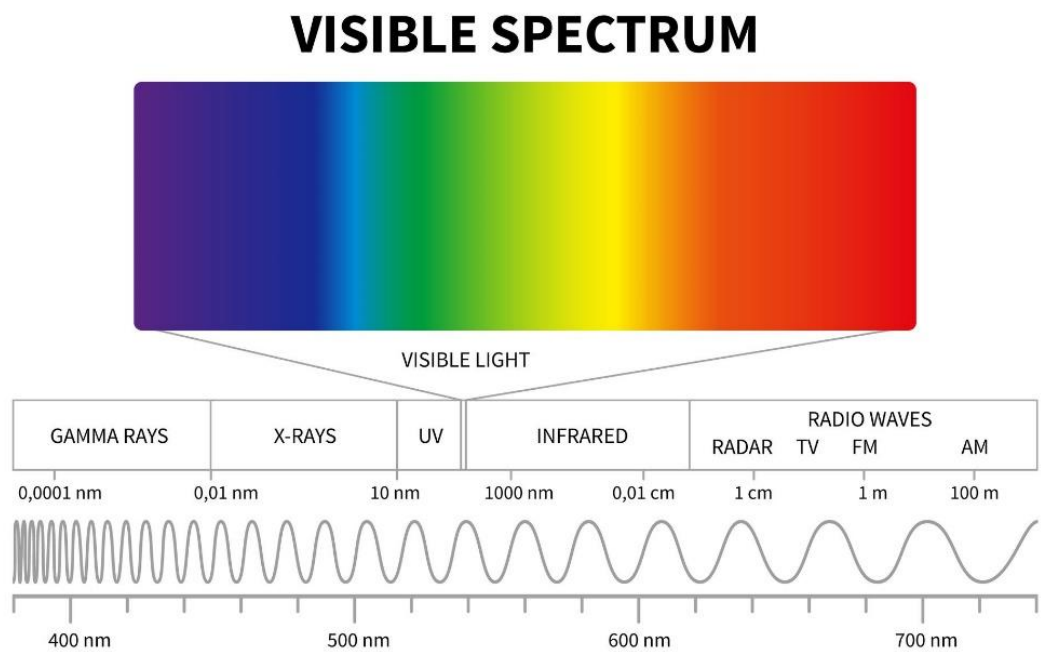
Table 3. Next-Generation Lithography Approaches and Challenges¹²⁰

NGL	Challenge(s)
Electron-beam lithography (a.k.a. e-beam direct-write lithography)	Masks: Electrons absorbed in the mask deposit energy, causing it to heat up and distort, resulting in overlay errors. The scattering with angular limitation projection electron-beam lithography program attempted to mitigate this, without success. Exposure tool: Throughput (wafers processed per hour) is limited due to beam current.
Extreme ultraviolet lithography	Optics: Figure and finish (shape and purity) requirements of mirrors are in the angstrom to sub-angstrom range.¹²¹ Essentially, the mirrors need to be atomically precise. Mask: Extremely limited defect tolerance. Atomic-level precision (30 angstrom) for multilayer coatings used to make masks. Light source: EUV light is difficult to generate reliably and absorbed by nearly all standard materials and environments (including glass and air).
Ion projection lithography	Light source: Reliable ion sources are large and expensive. Exposure Tool: Throughput (wafers processed per hour) is limited due to beam current.
X-ray proximity lithography	Optics: Lack of optics means changing image size to match previously printed layers is challenging. Mask patterning: Mask features must be the same size as the image being printed on wafer. Exposure tool: Sub-100nm manufacturing results in diffraction effects. Mitigation requires extremely precise management of mask-to-wafer gap (the mask and wafer can only be a few microns apart) and pre-distortion of mask patterns.

Source: Author's compilation.

The semiconductor industry made continuous progress on the light sources used in lithography as ever-diminishing feature size necessitated light sources capable of generating shorter wavelengths (Figure 7).

Figure 7. Visible Light Spectrum



Source: ASML, “Light and Lasers,” accessed June 26, 2024, www.asml.com/en/technology/lithography-principles/light-and-lasers.

Table 4 summarizes the commercial light sources and others that were considered for EUV lithography but ultimately abandoned in favor of the Cymer-developed carbon dioxide laser-produced plasma sources.

Table 4. Generations of Lithography Light Sources¹²²

	Light Source	Light Wavelength	Chip Feature Size
Commercial Use	Mercury arc lamps	G-line or “blue light” (436nm)	1 micron (1,000nm)
		I-line or “invisible ultraviolet light,” (365nm)	1 micron–220nm
	Excimer lasers	Krypton-fluoride based (248nm–150nm)	220nm–80nm
		Argon-fluoride based (193nm)	80nm–38nm
	Carbon-dioxide laser	Tin-based plasma (13.5nm)	38nm–2nm (or lower)
R&D Use	Free electron laser	1mm–.01nm	N/A (not used in chip fabrication)
	Steady-state microbunching	Unknown	N/A (not used in chip fabrication)
	Compact synchrotron laser source	Unknown	N/A (not used in chip fabrication)

Source: Author’s compilation.

Appendix 3. Comparing EUV Publication Intensity between SPIE and Scopus/Web of Science, 1994–2017

